



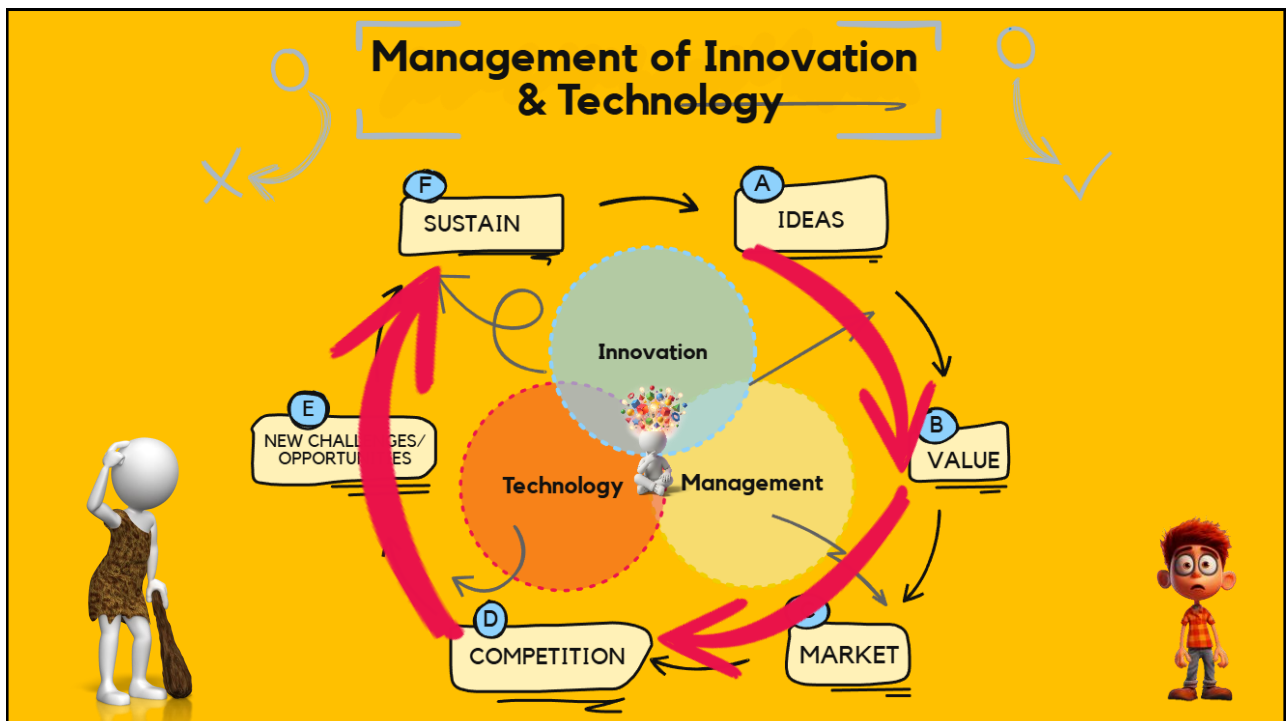
ISE549 Management of Innovation and Technology

Creativity, Innovation & the WOW effect

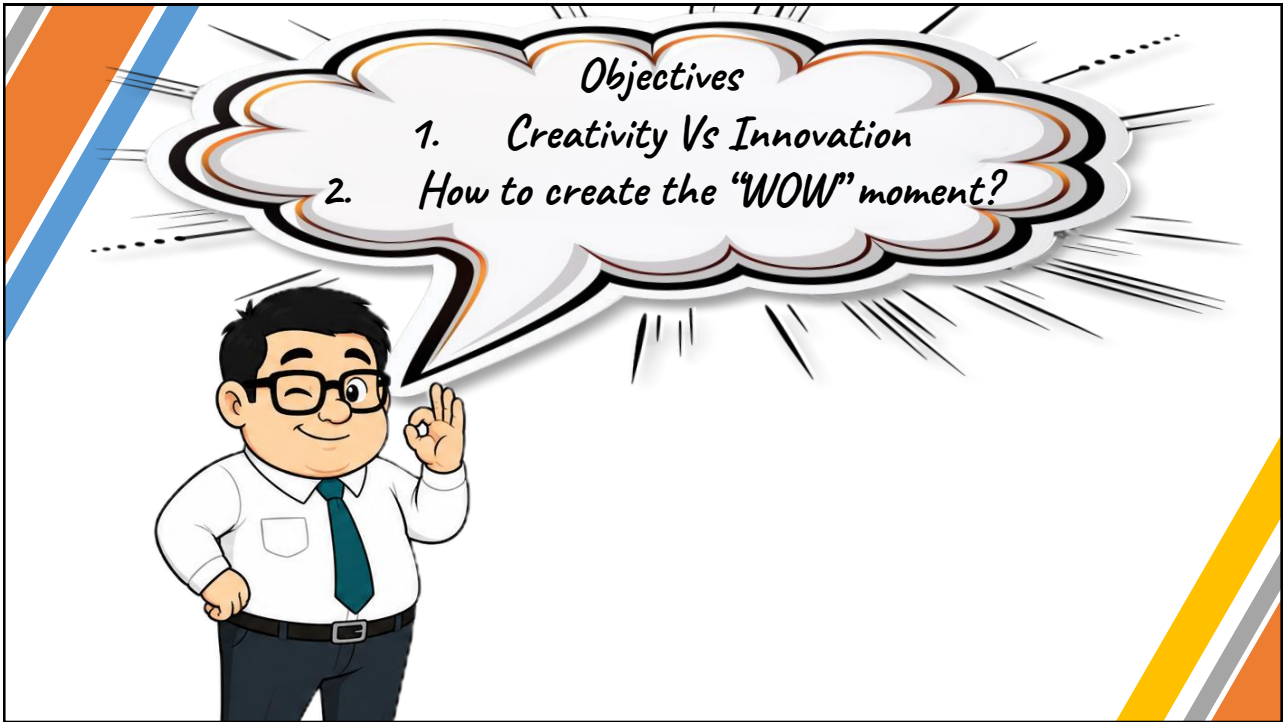
Prof. Eric Tsui

Educational Research & Technologies Centre
The Hong Kong Polytechnic University, Hong Kong

123

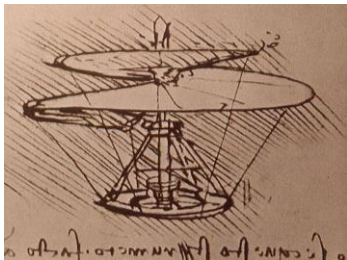


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What is Creativity?



- **Creativity** is a mental and social process involving the generation of new ideas or concepts, or new associations of the creative mind between existing ideas or concepts. **Creativity** is fuelled by the process of either conscious or unconscious insight.
- [Steps To Innovation And Creative Thinking](#)
- Need to explicitly distinguish between **creativity** and **innovation**.
- **Creativity** is typically used to refer to the act of producing new ideas, approaches or actions, while **innovation** is the process of both generating and applying such creative ideas in some specific context.
- [Edward de Bono on innovation vs creativity how's it different](#)

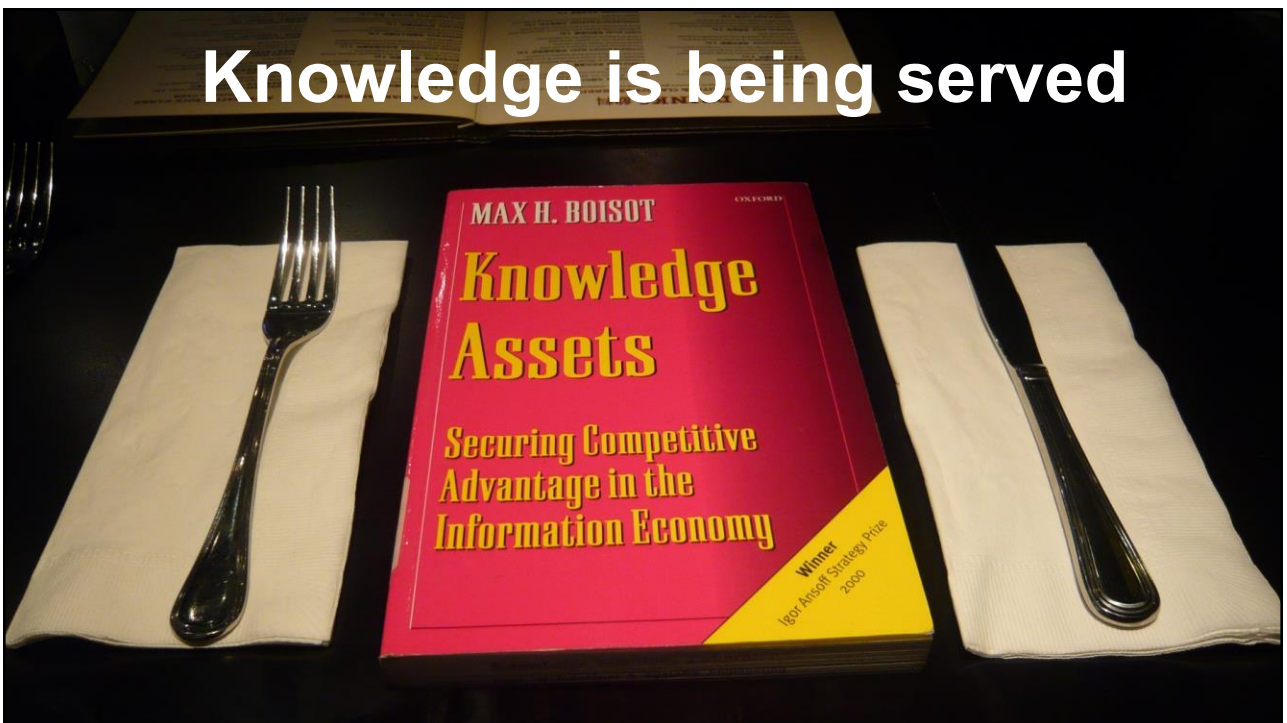


Source: Simon Li, 2018

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What is the design rationale for this seating?

How to ask for a glass of water?
Have a creative mindset

Care to have a discussion in a shower?



131

A fine line between creative & madness



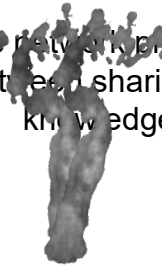
132

The new paradigms of Creativity



The new paradigm of creative industries is settling as a collective process that should be share through the net, communities of practise, brainstorming, etc.

In the network phenomenon, the gap between sharing and protecting knowledge is reducing.



The diffusion of knowledge is both a top-down and bottom-up process that moves through the borders.

Source: Simon Li, 2018

135

Technological support for the three Pillars of Creativity



Expertise

- technical, procedural & intellectual knowledge
- Required tools

Creative thinking skills

- how flexibly and imaginatively people approach problems

Motivation

- especially intrinsic motivation

Technology

- Allows easy access to information and resources

Technology

- Stimulates imagination
- Change
- AI helps think

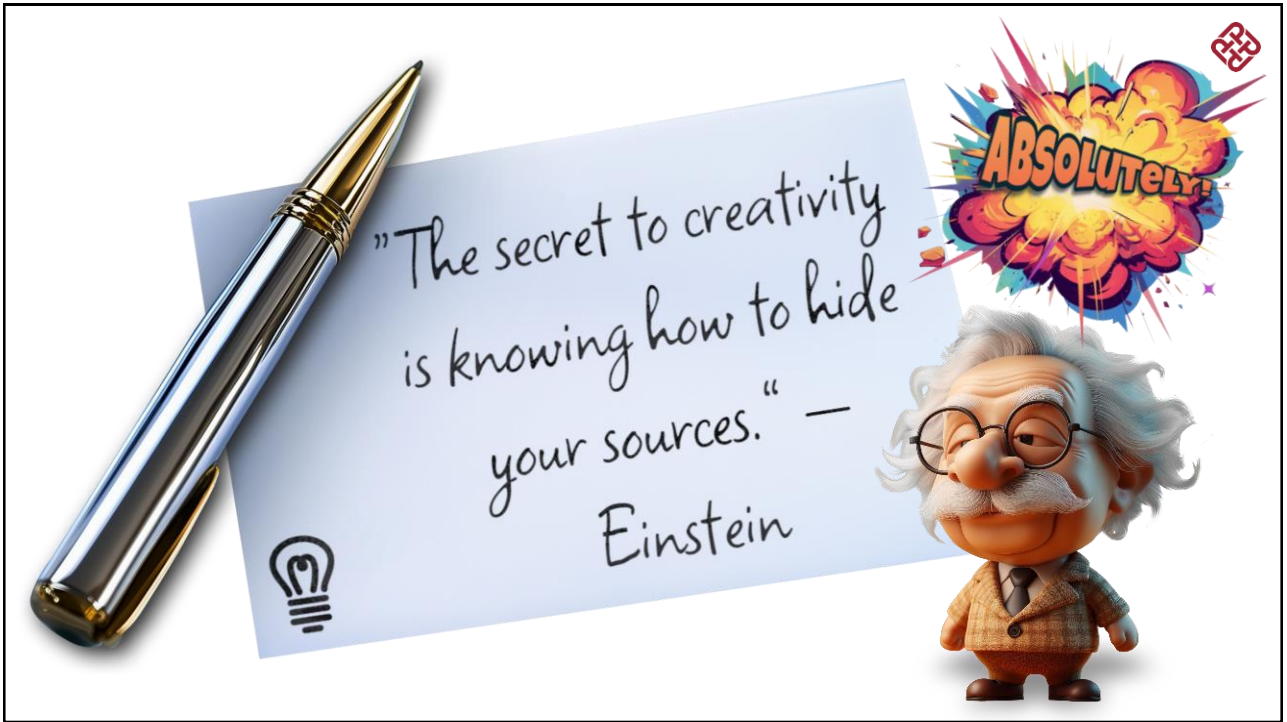
Technology

- Allows for "mass motivation"

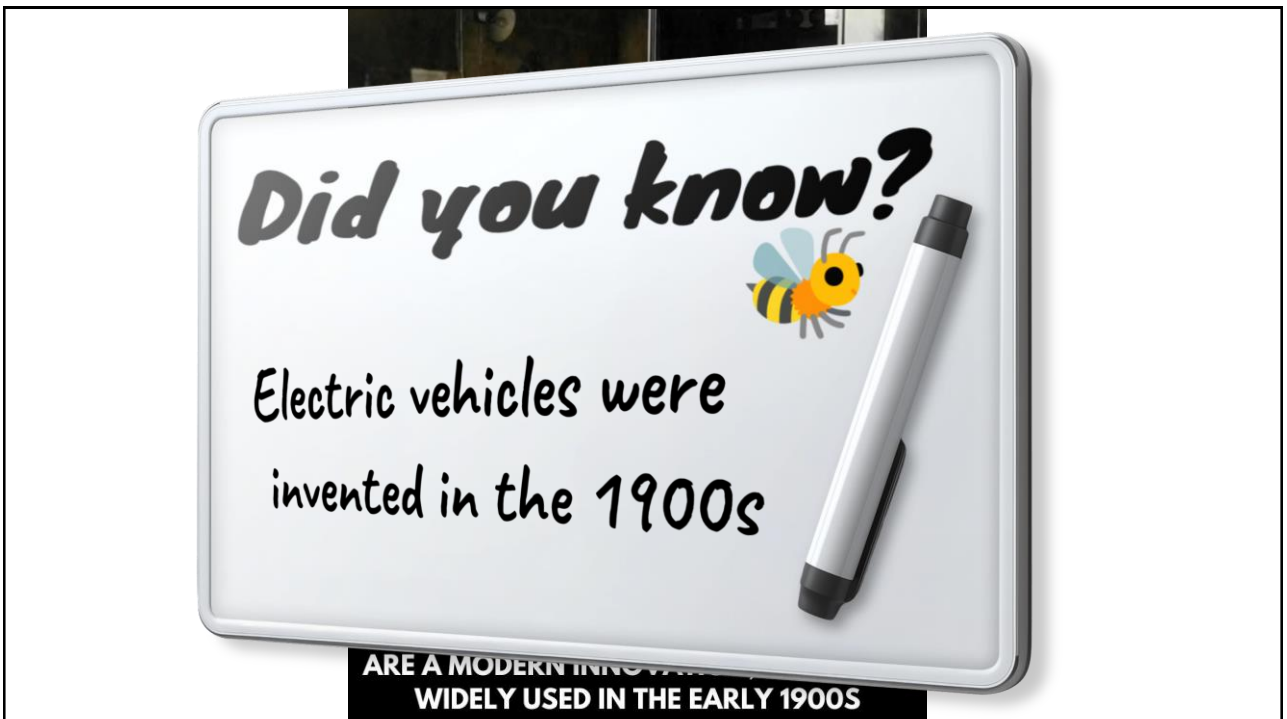


Source: Simon Li, 2018

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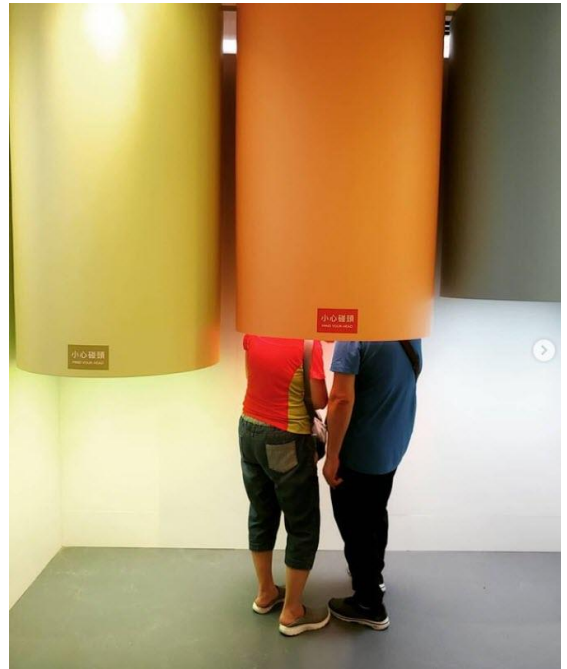


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144

The “Cone of Silence” in Get Smart TV series in 1960s



145

Creative by copying someone else's ideas and improve...



146

How could this be a 1965 tech?



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Creativity vs. Innovation

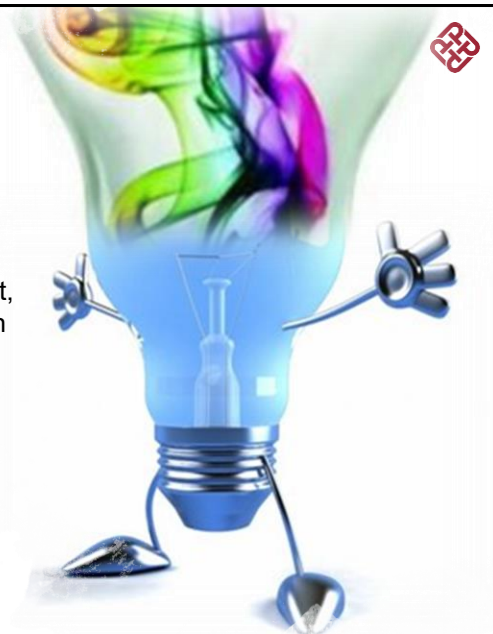
Creativity is giving life to something which does not exist before and it is just the first step of innovation.

Innovation is the application of creative ideas to a new product, a new productive process or a new kind of organization in such a way to have an economic return.



NOT ALL CREATIVE ACTS PRODUCE AN INNOVATION

(Usually, creative works can have a high failure rate from perspectives of the management)



Source: Simon Li, 2018

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Definition of Innovation



Innovation is conceived as a means of changing an organization, either as a response to changes in the external environment or as a pre-emptive action to influence the environment. Hence, innovation is here broadly defined to encompass a range of types, including new product or service, new process technology, new organization structure or administrative systems, or new plans or program pertaining to organization members [DAM 98].

151

Innovation represents the core renewal process in any organization. Unless it changes what it offers the world and the way in which it creates and delivers those offerings, it risks its survival and growth prospects [BES 05].



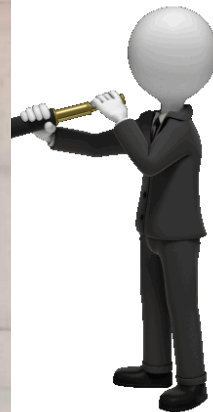
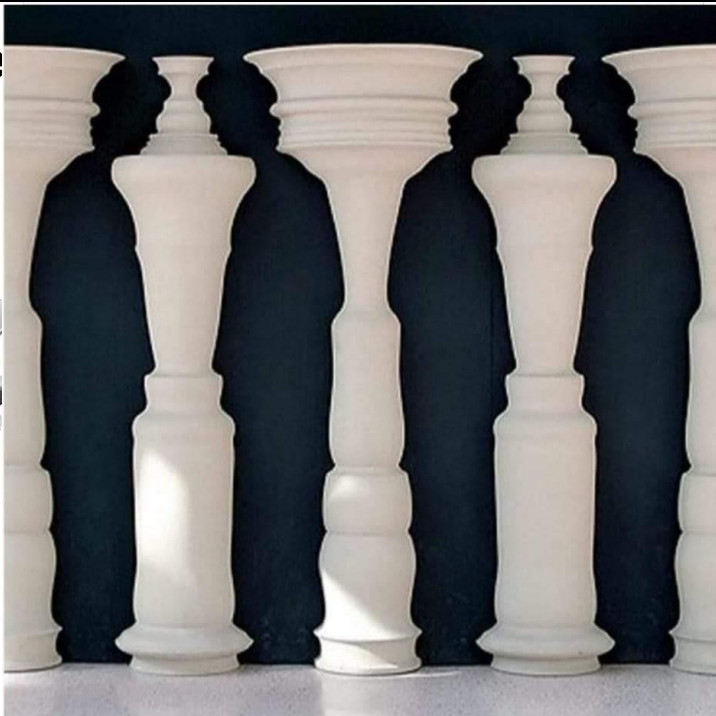
Innovation is the creation of new knowledge and ideas to facilitate new business outcomes, aimed at improving internal business processes and structures and to create market driven products and services. Innovation encompasses both radical and incremental innovation [PLE 07].

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Creative



157

Creative



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Creative or Innovative?



159

Creative or Innovative?



160

Creative or Innovative?



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Disproportionate in scale



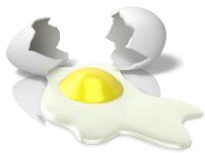
167

Disproportionate in scale



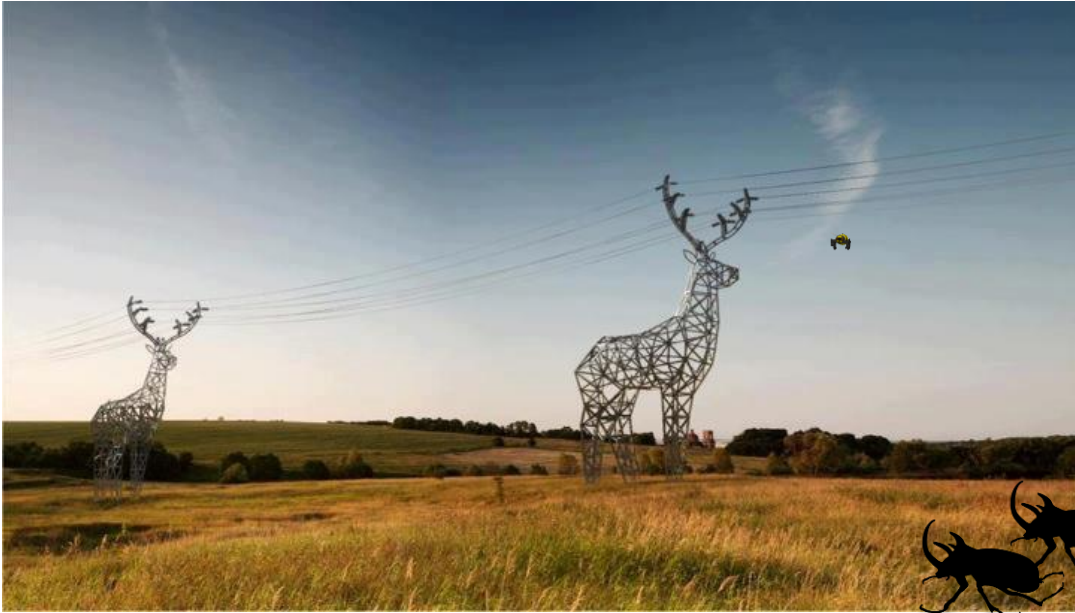
169

Creative or Innovative?



170

Creative or Innovative?



171

What is the innovation?

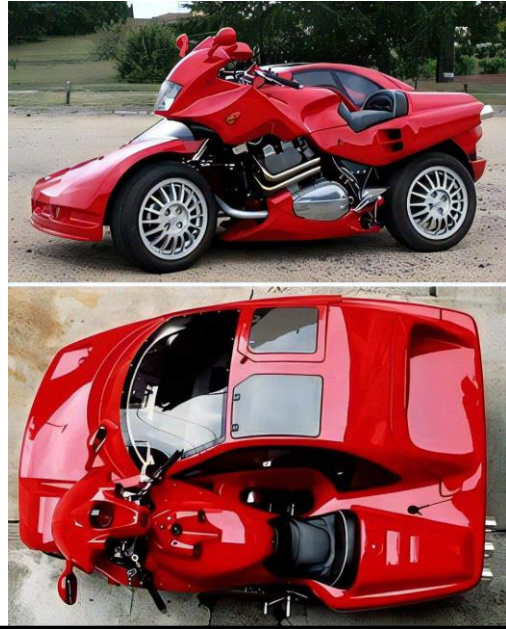


Source: Nick Chung, PolyU, 2018

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Innovation in hybrid mode



176



PC adaptor with variable voltage



177

Foldable monitor



178

What is the innovation?



Source: Anthony J. James, 2021

179

Extendable garage



Retractable sunshades



182

Pushbike and Bicycles



184

ELIDE Fire Extinguishing Ball



186

Traffic lights



187

“Glow-in-the-dark” lane lines with phosphorescent pigments



188

Looking/Knowing what's ahead



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Innovations in lift design



Paternoster Elevator, or continuous elevator, does not have doors and never stops

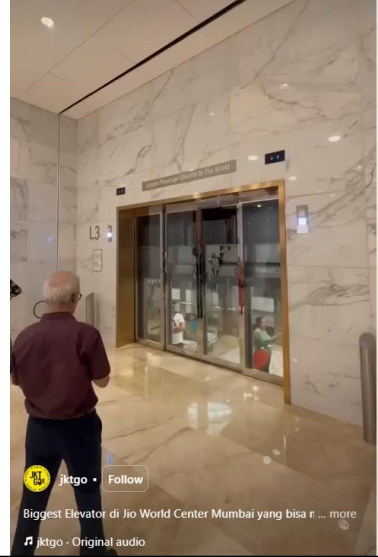


callitdesignco • Follow

An 1860 Innovation - This video takes you into the fascin ... more
callitdesignco - Original audio

**Challenge
the norm!**

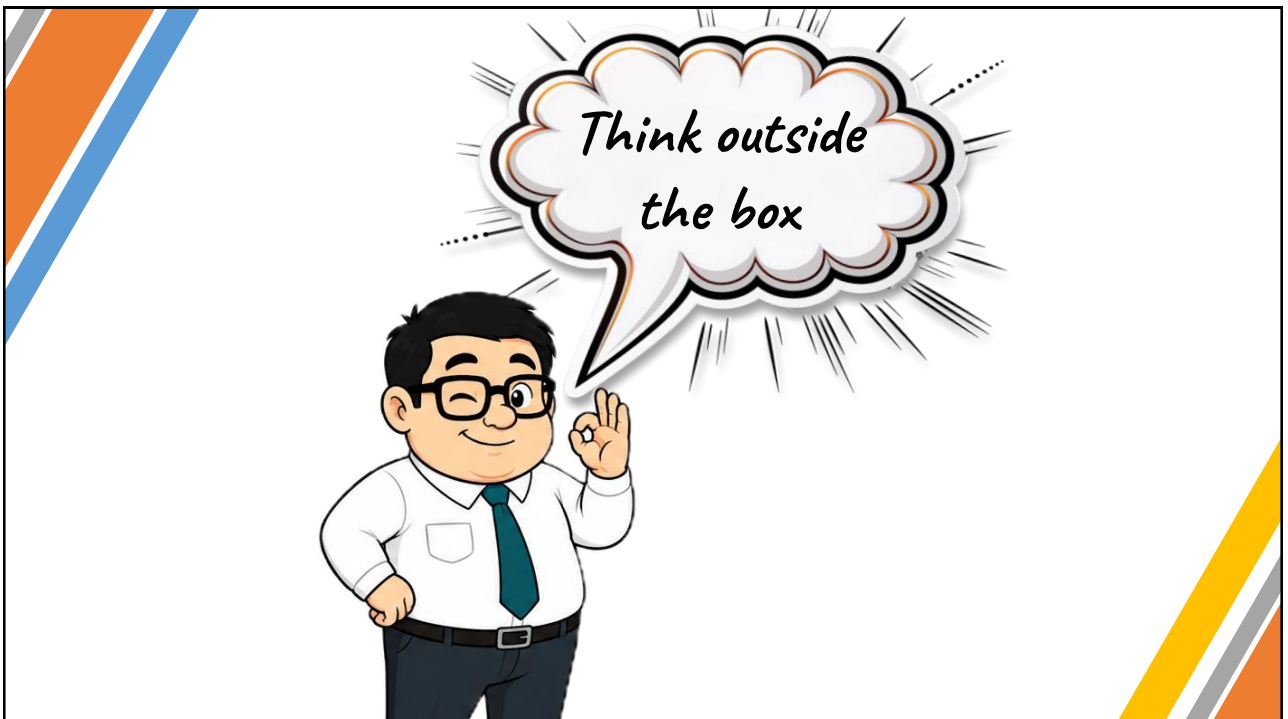
This is the largest passenger elevator in the world



jkto • Follow

Biggest Elevator di Jio World Center Mumbai yang bisa r ... more
jkto - Original audio

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191

Breaking a rule: Does a wheel always need to be even and in a round shape and ?



194

Which parts belong to the advertisement?



195

Chilli hot pot for a meal



Chilli hot pot for a *bath*



196

Clamping machine in a fun parlour



197

Innovations with the “toilet seat”



In Japan, a sink is built above the toilet tank for hand washing. The water drains into the toilet tank to be used for the next flush, and saves millions of gallons of water each year.



Japan Develops a **Smart Toilet** That **Analyzes** Your **Waste** and Sends a **Personalized Health Report**



SINGAPORE USES ROBOTIC SWANS TO MONITOR LAKE WATER QUALITY BY MONITORING PH, OXYGEN AND POLLUTANTS

199

Innovations with the “toilet seat”



203

“New design” for toilets



204

Social Innovation: A corporate retreat in Australia

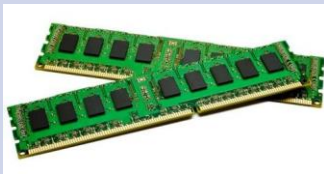


205

Social Innovation: CEO sleeping out on streets



206

Item 1	Item 2	Creative design
		
		
		

211

A mini projector



213

***What
sparks
the “Ah-
ha” s
and the
“WOWs”
in us?***



- Increased Speed
- Reduction in Size
- Less Weight
- Lowered Cost
- More Functions
- Unusual shape
- Ease of access/use
- Saved time
- Automation
- Embedded intelligence
- Removed a “pain point” / intermediary
- Something rare, illusion, something very few people can do or contrary to our belief e.g. an unusual pattern, disproportionate size
- Refreshing experience



217

***What sparks
the “Ah-ha” s
and the “WOWs”
in us?***



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- Refreshing experience



218



- What if a customer subscribes to your business offerings?
- What if we need new core competencies?
- What if what your customer value wasn't based on money?
- What if you look at the problem from someone else's point of view?
- What if you partner with a competitor?
- What if how people worked in your company changed?
- What if nobody is allowed to travel?
- What if customers can also sell for you?
- What if price is too high?
- What if it the product needs customization for personal experience?
- What if you didn't have any employees?

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Opportunities for sparking innovations

- Unexpected occurrences
- Incongruities
- Process needs
- Industry and market changes
- Demographic changes
- Changes in perception (e.g. get out of our comfort zone)
- Resolves a pain point
- New knowledge (internal & external, absorptive capacity)



Source: HBR Inspiring and Executing Innovation, 2011

220



222



223



224

Customer Engagement

REI – closing their stores on the busiest shopping day: Outdoor equipment retailer REI had begun closing its doors on Black Friday, traditionally one of the busiest shopping days of the year. They claim they are doing this to Eddie their customers to actually get outdoors and use their equipment, rather than queuing for discounted material goods. REI – closing their stores on the busiest shopping day: Outdoor equipment retailer REI had begun closing its doors on Black Friday, traditionally one of the busiest shopping days of the year. They claim they are doing this to Eddie their customers to actually get outdoors and use their equipment, rather than queuing for discounted material goods. (Source: ideatovalue.com/inno/nickskillicorn/2019/07/ten-types-of-innovation-30-new-case-studies-for-2019)



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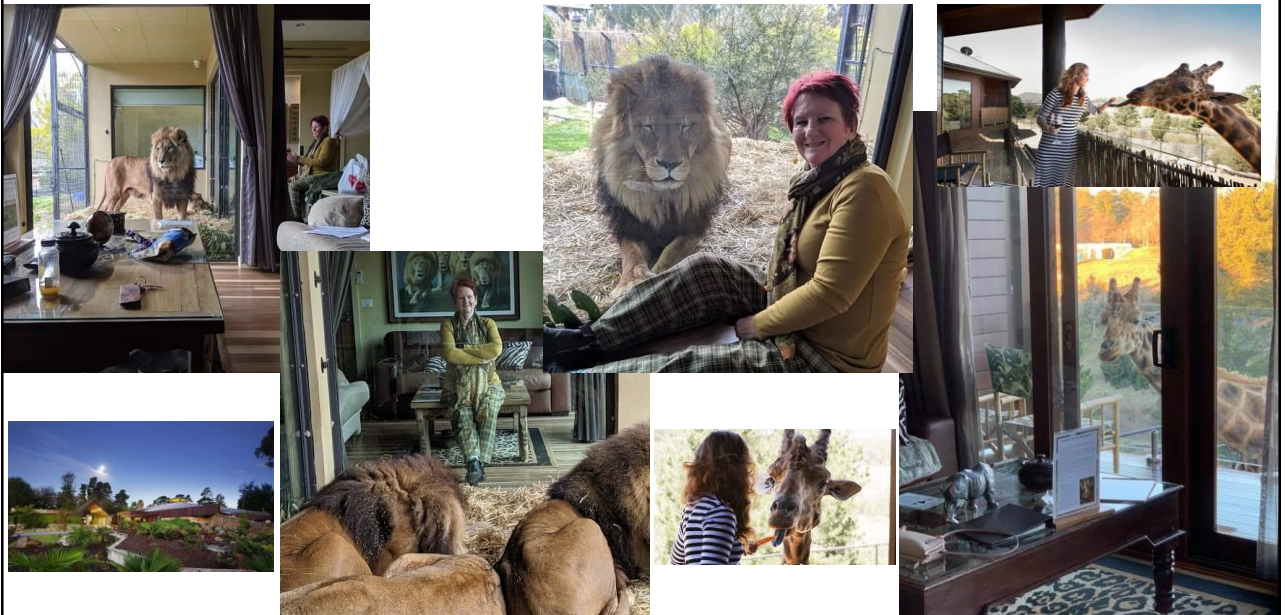
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Break the glass and you can take the cash



228

Jamala Wildlife Lodge, ACT, Australia



229



There is a '**Reverse zoo**' in china that puts the **visitors inside cages** and allows the **animals to roam free.**



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231

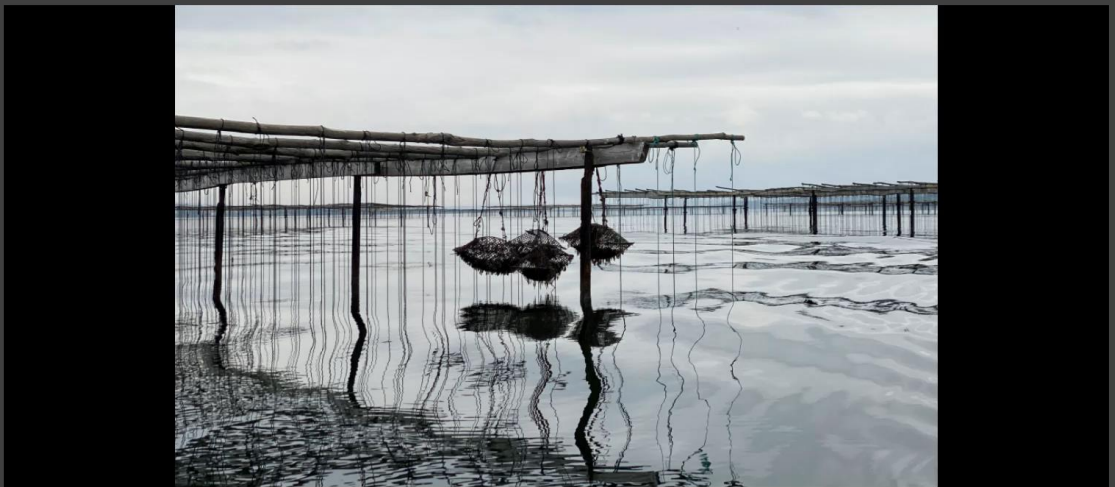
Worker Smarter not harder



232

Innovation as a smart solution

"You've won your weight in oysters!" - note in shells stops thieves



4/4

A general view shows an oyster farm, near Leucate, southern France, December 23, 2021. Picture taken December 23, 2021. REUTERS/Alexandre Minguez



233

Innovation as a smart solution



Who said you have to ride your bike? "Fly on your bike" instead



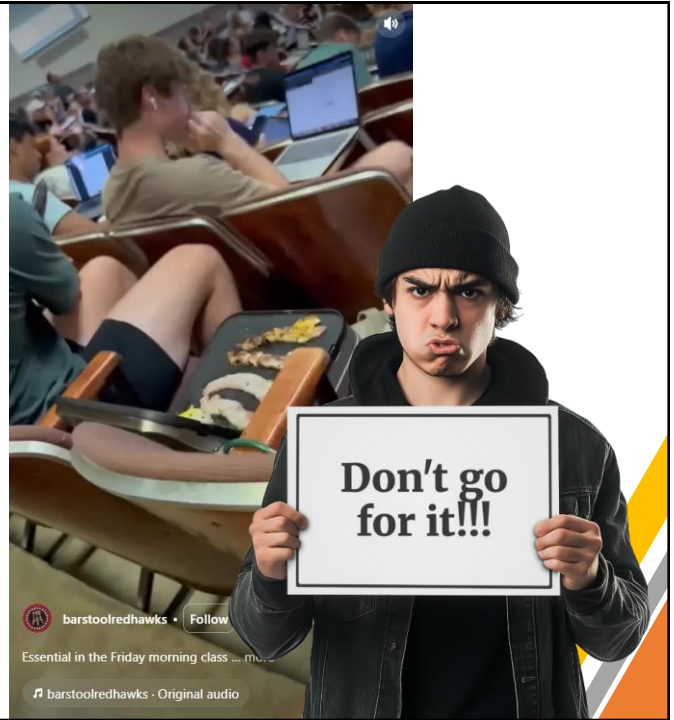
234

Innovation that should not be copied!!!



236

Innovation
that should
not be
copied!!!



237

Innovation
that should
not be
copied!!!



238

Innovation
that is not
worth to
build!!!



239

7 COMPANIES THAT WERE SUED FOR CAUSING CANCER

DavidAvocadoWolfe

Irresponsible
& unethical
practice in
innovations

**OLD SPICE DEODORANT**

Use: Benzene

**JOHNSON & JOHNSON**

Sued for: Use of asbestos

**STARBUCKS COFFEE**

Sued for: Acrylamide

**BANANA BOAT**

Sued for: Benzene

**COLGATE-PALMOLIVE**

Reason: Lead

**CHEERIOS CEREAL**

Reason: Glyphosate

**CHOBANI YOGURT**

Sued for: Use of phthalates



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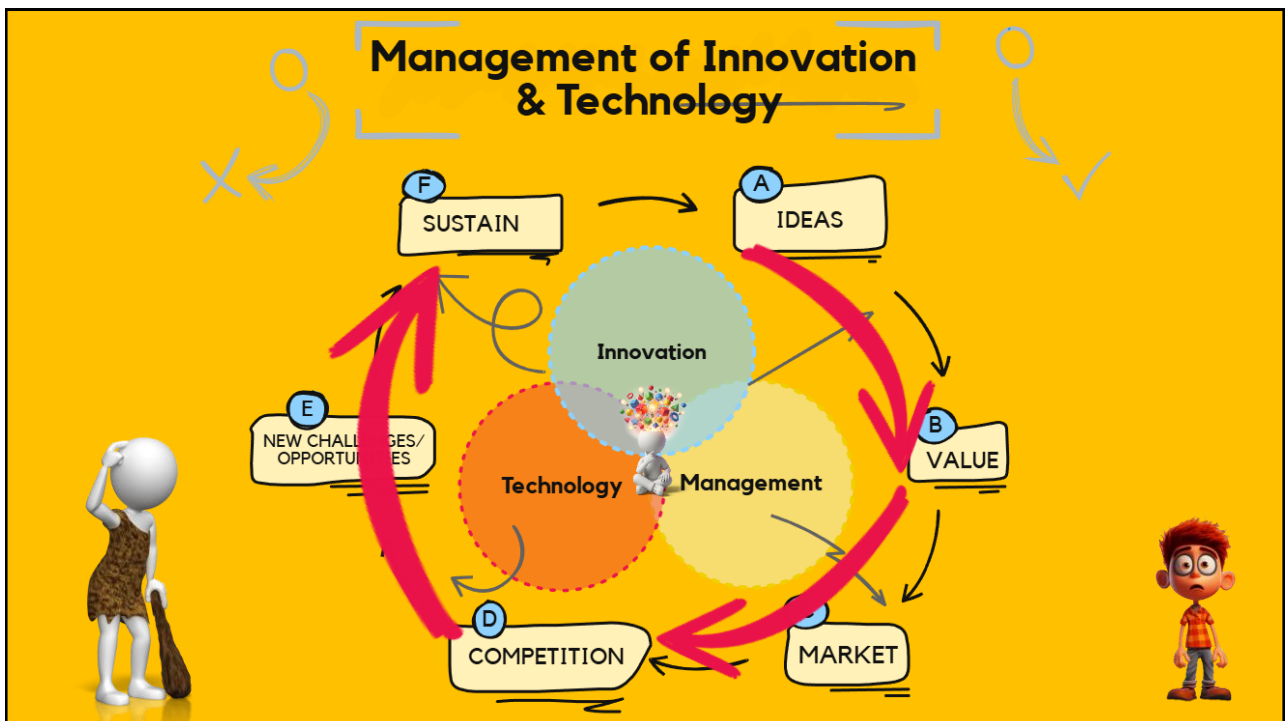
ISE549 Management of Innovation and Technology

Theories & types of Innovation, innovation process & systems

Prof. Eric Tsui

Educational Research & Technologies Centre
The Hong Kong Polytechnic University, Hong Kong

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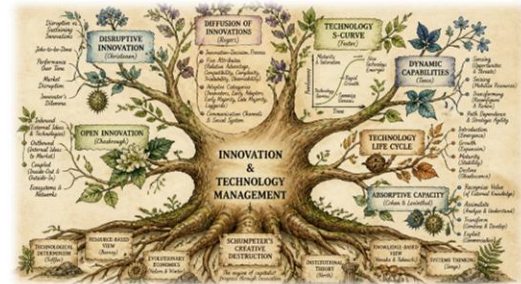


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Innovation

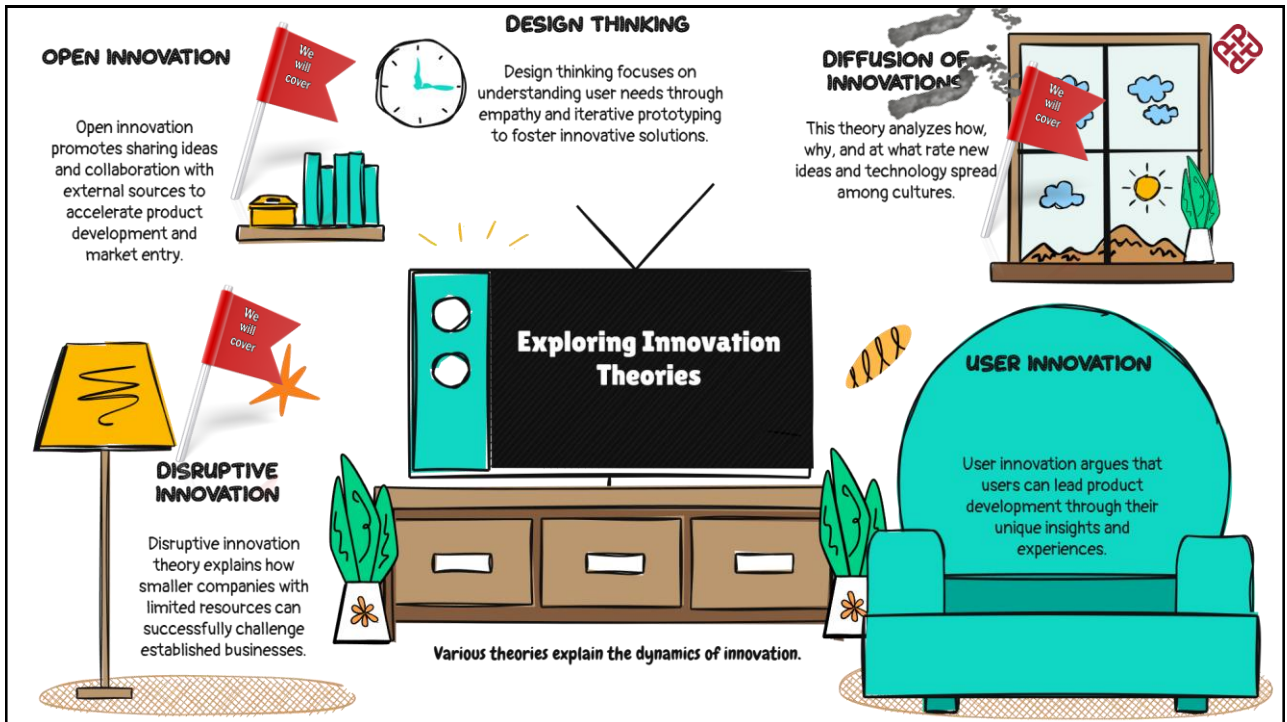
- Theories of Innovation
- Innovation Process
- Types of Innovation



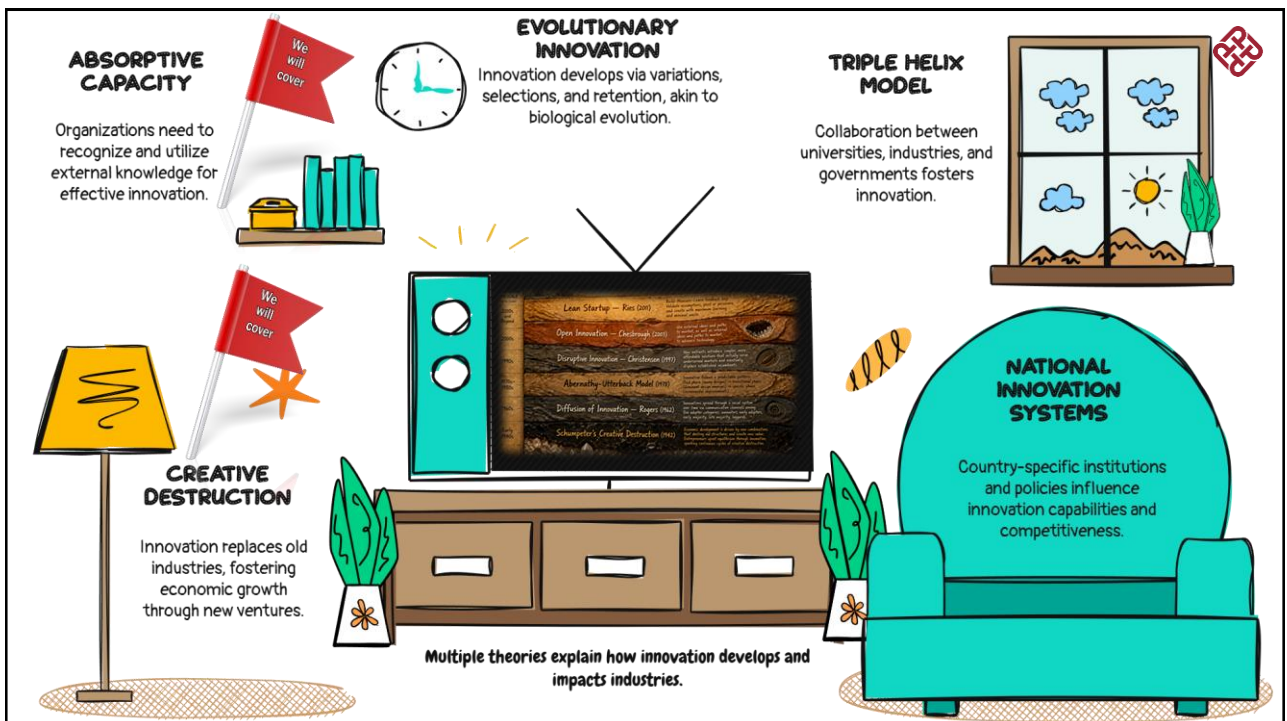
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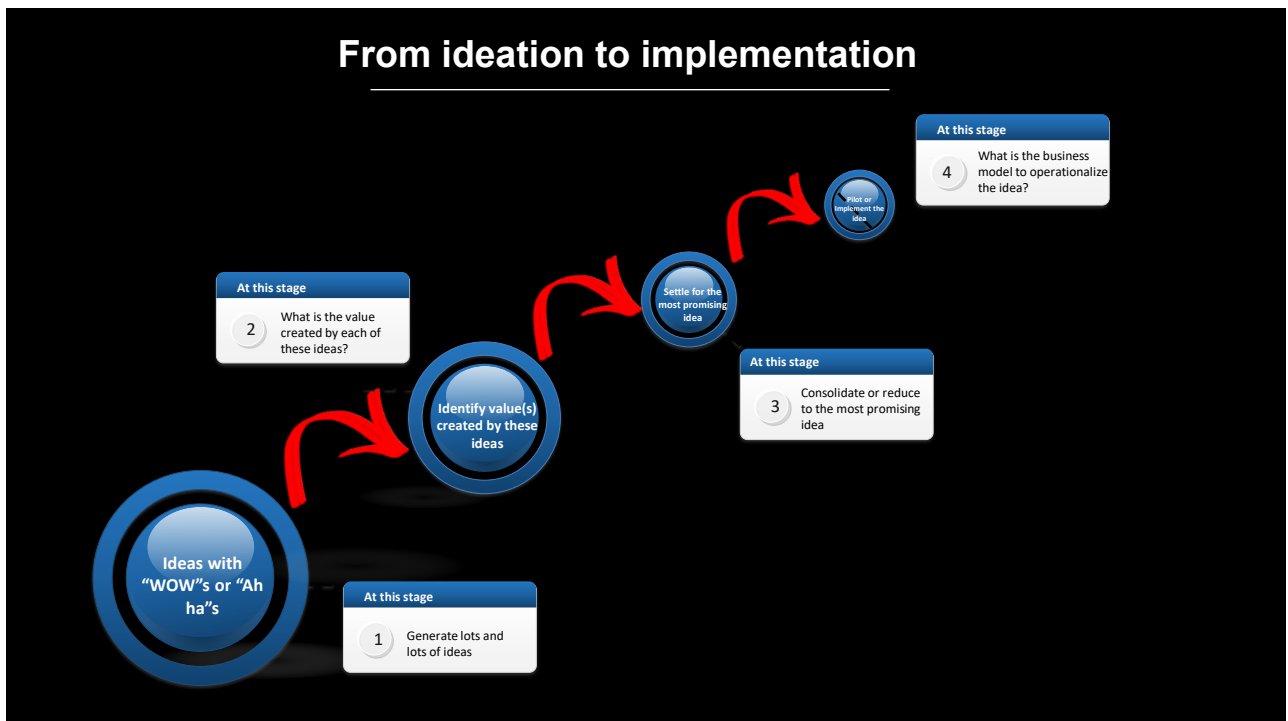
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**What
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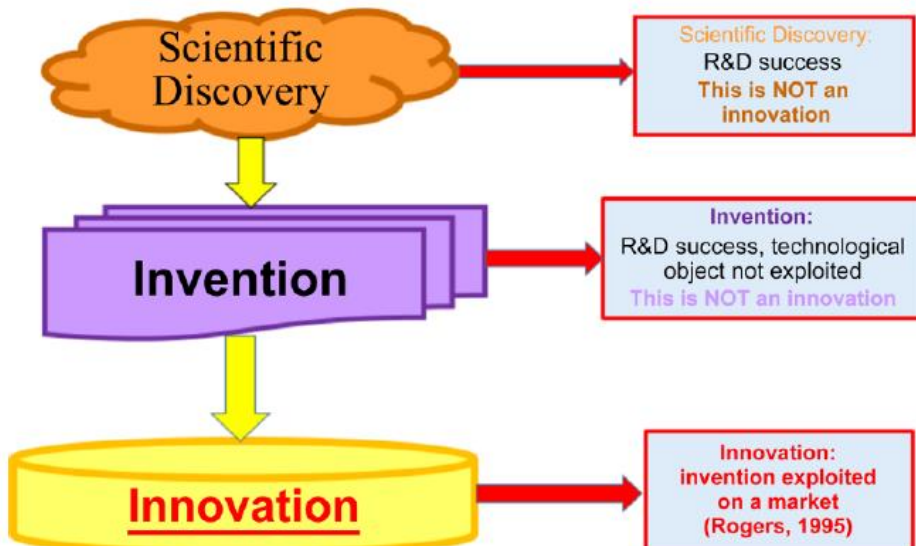


- Increased Speed
- Reduction in Size
- Less Weight
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- Automation
- Embedded intelligence
- Removed a “pain point” / intermediary
- Something rare, illusion, something very few people can do or contrary to our belief e.g. an unusual pattern, disproportionate size
- Refreshing experience



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Innovation process: From discovery to innovation

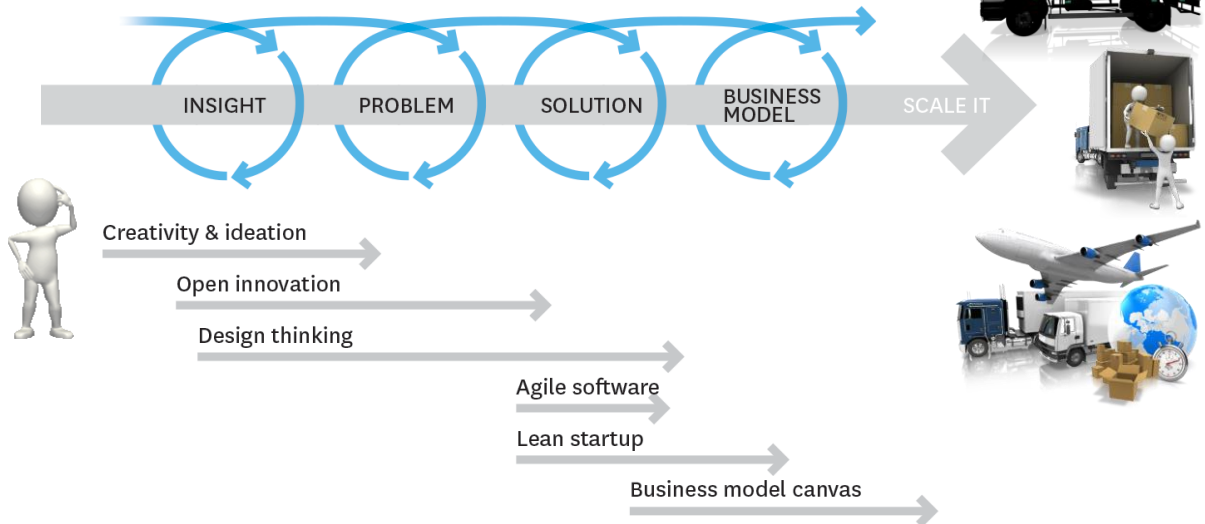


Source: O. de Saint Julian, The Innovation Ecosystem as a source of Value Creation, Wiley. 2022.

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AN END-TO-END INNOVATION PROCESS

Adapting the tools honed by start-ups.



SOURCE NATHAN FURR AND JEFF DYER

HBR.ORG

Source: Furr & Dyer (2014)

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Key Elements of an Innovation Ecosystem

Understanding the core components driving innovation success.

RESEARCH INSTITUTIONS

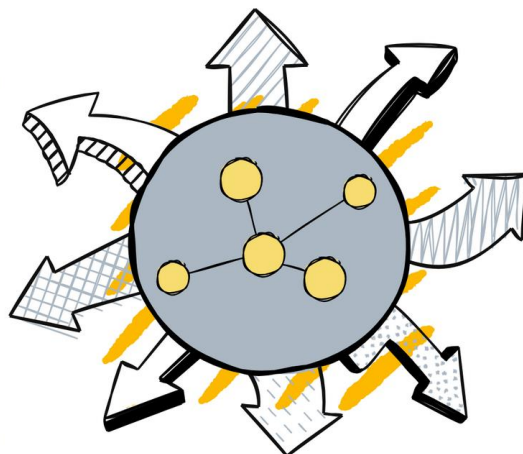
Research institutions are foundational for innovation, providing critical knowledge and technological advancements that drive the development of new ideas and solutions.

INVESTORS

Investors fund innovative projects and startups, providing necessary resources and financial backing crucial for growth and sustainability in the market.

COLLABORATIVE NETWORKS

Collaborative networks foster partnerships between various entities, facilitating knowledge exchange and resource sharing to boost innovation activities.



ENTREPRENEURS

Entrepreneurs play a vital role as they identify market needs and leverage opportunities to transform innovative ideas into viable businesses.

GOVERNMENT POLICIES

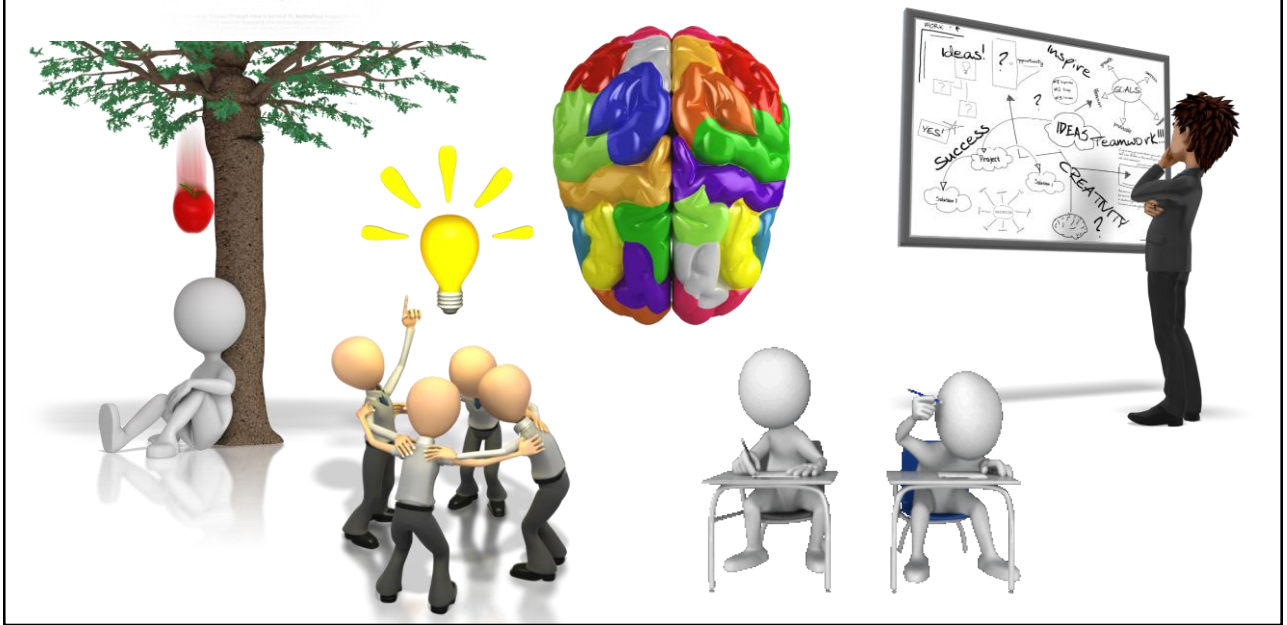
Supportive government policies can enhance innovation by creating favorable regulatory environments, offering grants, and promoting research and development initiatives.

TALENT POOL

A skilled talent pool is essential for driving innovation, as it equips organizations with the necessary expertise to develop and implement new ideas.

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TYPES OF INNOVATION



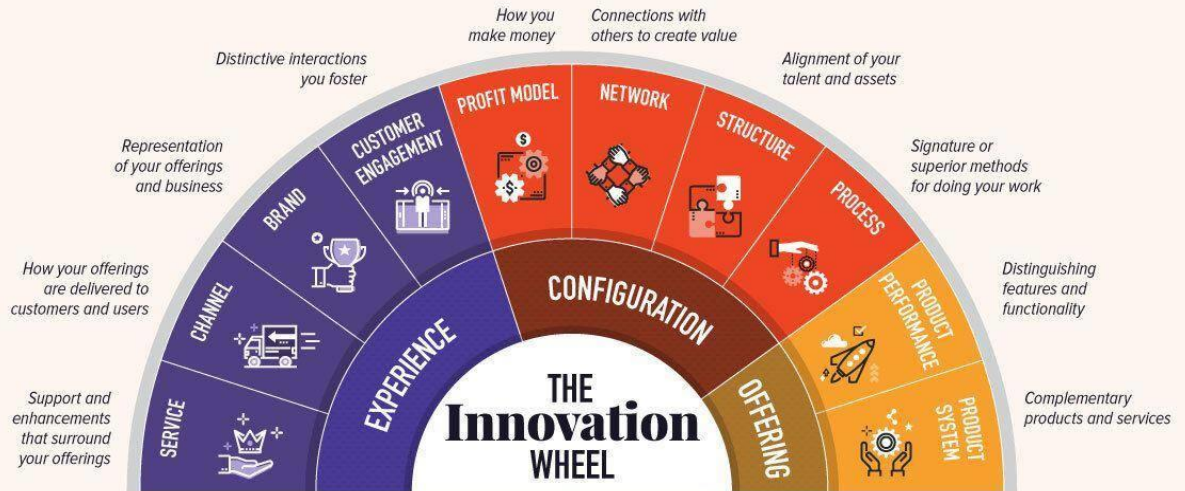
424

Two opposing schools



426

UNDERSTANDING THE 10 Types of Innovation

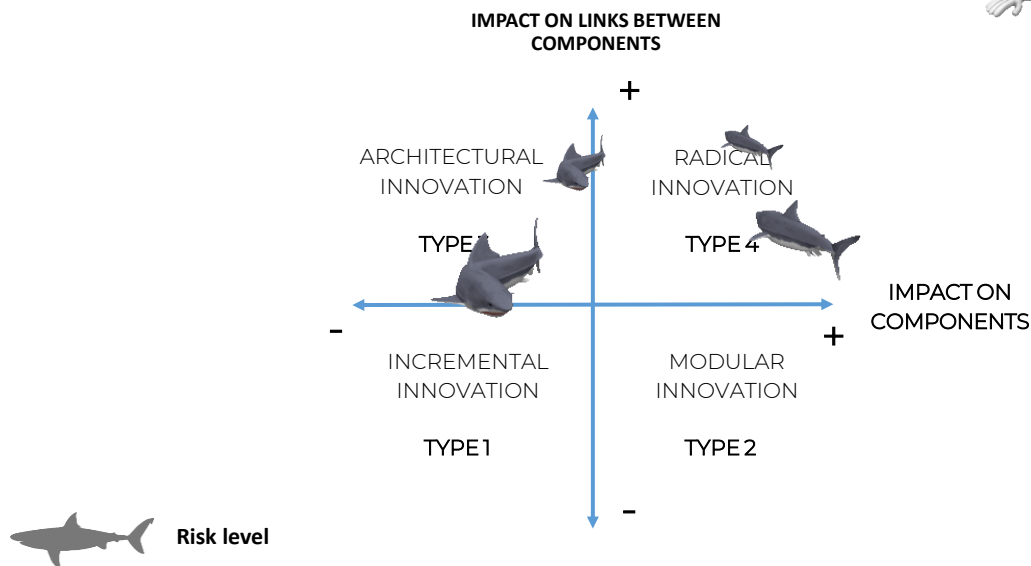


<https://doblin.com/ten-types>

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FOUR GENERIC TYPES OF INNOVATION

Source: Henderson and Clark's four types of innovation (1990)



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Modular, Architectural & Incremental innovations

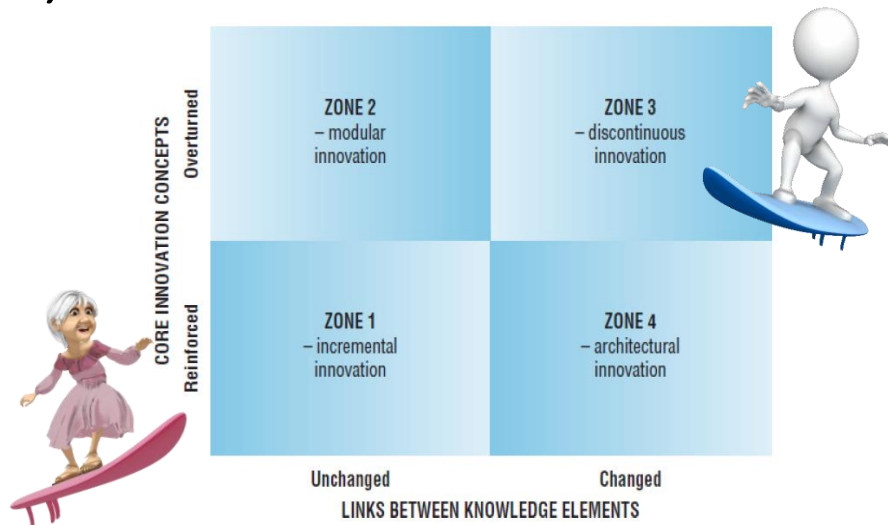


FIGURE 1.5 Component and architectural innovation

Source: J. Tidd and J. Bessant, *Managing Innovation*, 5th edition, Wiley, 2013.

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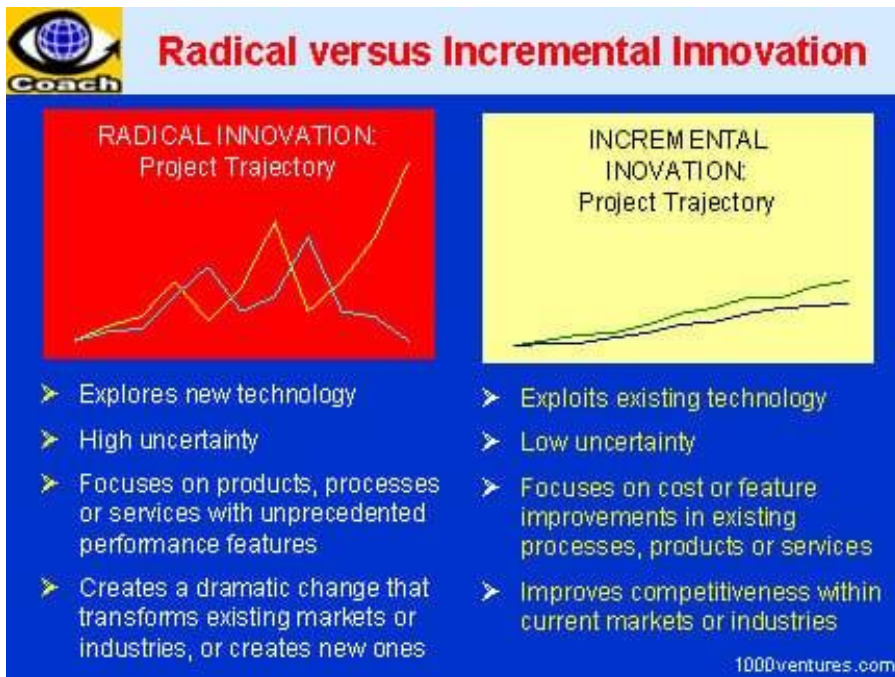
Radical Innovation Vs Incremental Innovation

- The radicalness of innovation is sometimes defined in terms of the **extent/speed of change** and/or the **risk**
- The radicalness of an innovation is **relative**, and may change over time or with respect to different observers



Source: M.A. Schilling, *Strategic Management of Technological Innovation*, 6th Edition, McGraw Hill, New York, 2020.

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component (or modular) innovation

An innovation to one or more components that does not significantly affect the overall configuration of the system.

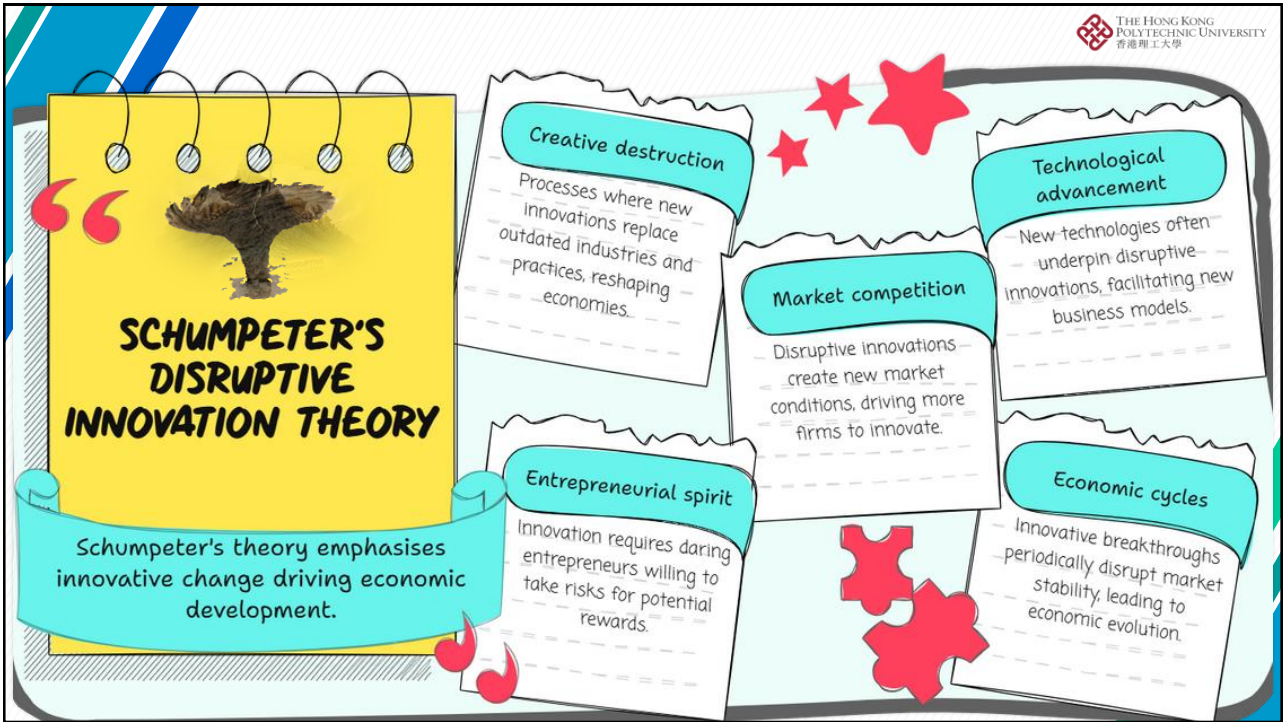
architectural innovation

An innovation that changes the overall design of a system or the way its components interact with each other.

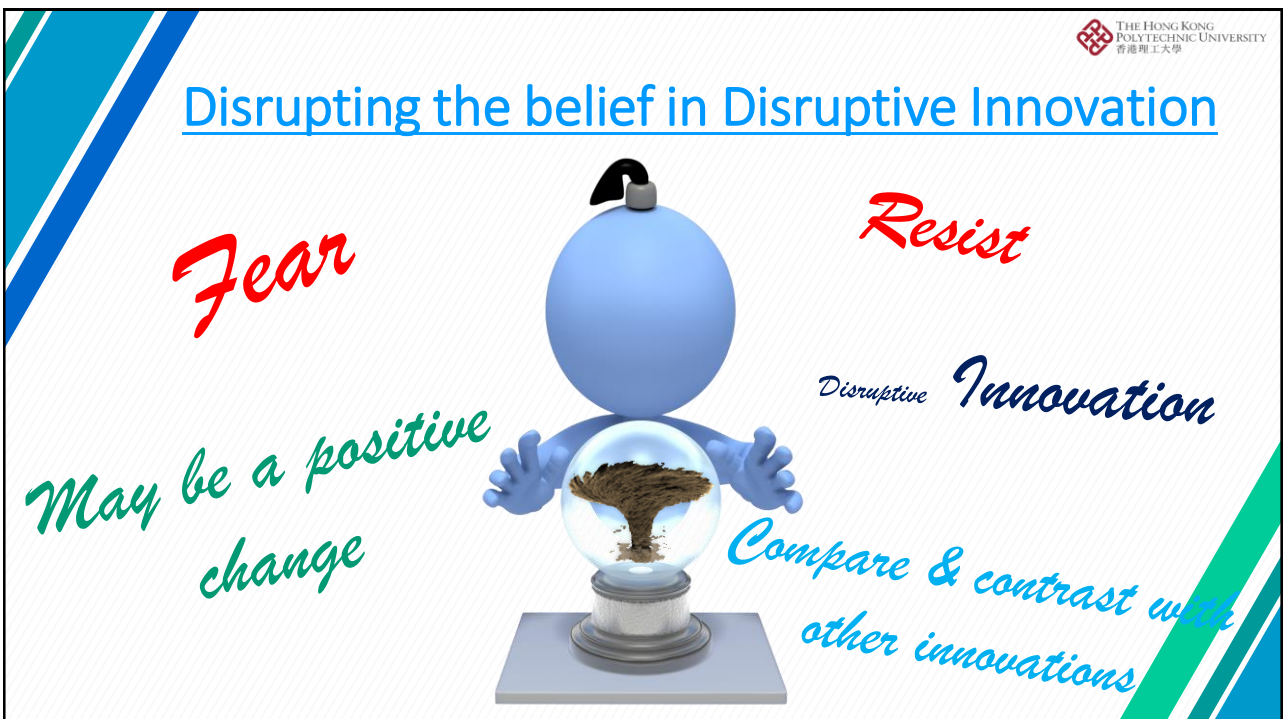


Source: M.A. Schilling, Strategic Management of Technological Innovation, 6th Edition, McGraw Hill, New York, 2020.

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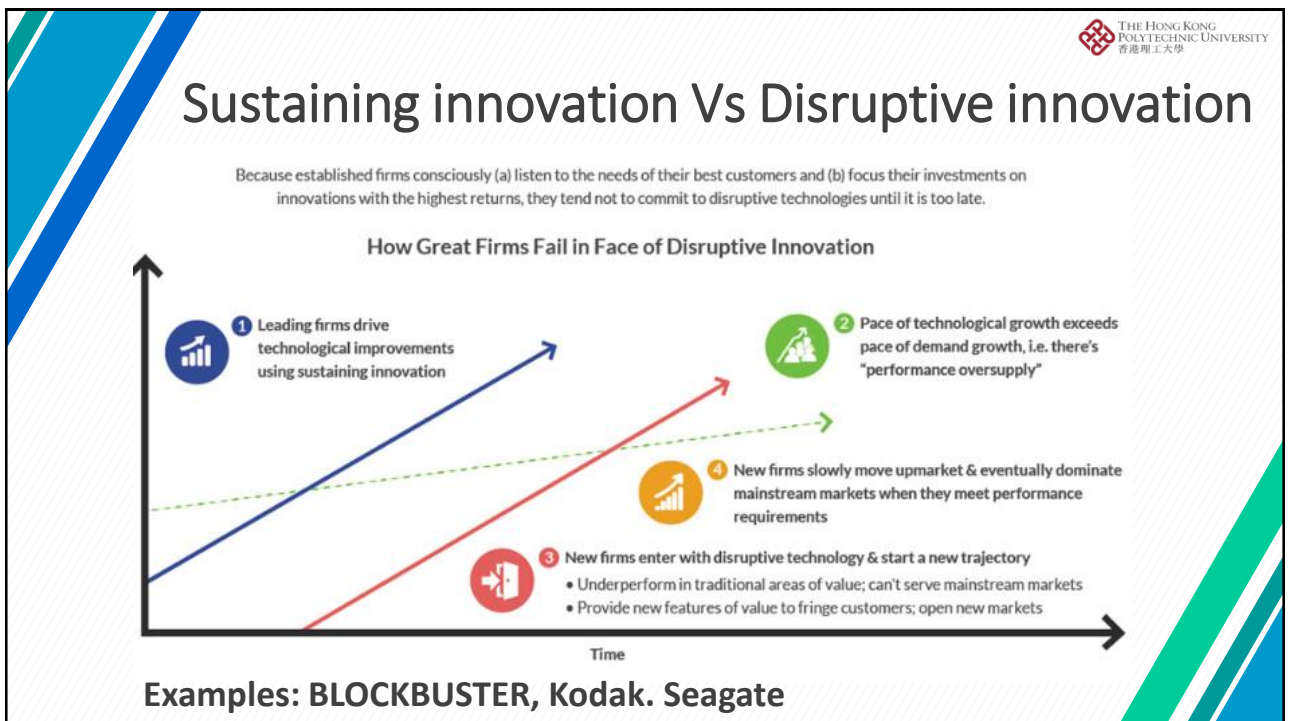
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Budget airlines disrupting traditional airlines

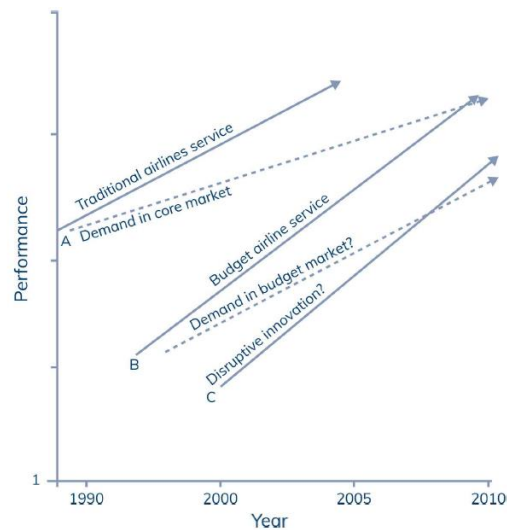
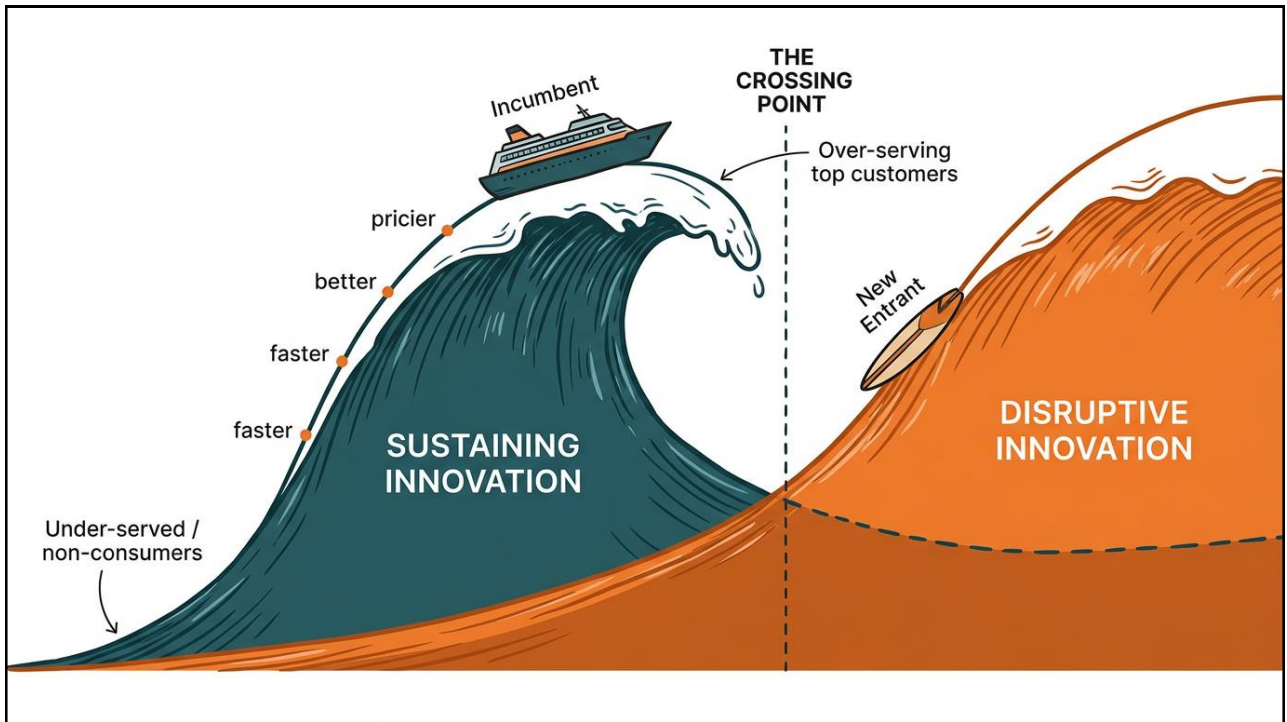


Figure 4.3 The disruptive innovation model

Source: Christensen (1993). Adapted for the airline industry instead of the hard-disk-drive industry.

Source: J. V.D, Ende, Innovation Management, Macmillan International, 2021.

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THE HONG KONG
POLYTECHNIC UNIVERSITY
香港理工大學

Sustaining innovation Vs Disruptive innovation

(Interview with Prof Jose Alves, City University of Macau)



Tsui Yue Hong Eric



商學院
Faculty of Business



Replay at <https://www.youtube.com/watch?v=e6cuB1hYkaQ&feature=youtu.be>
Click [here](#) to read the article

444

Product style Vs technological uncertainty

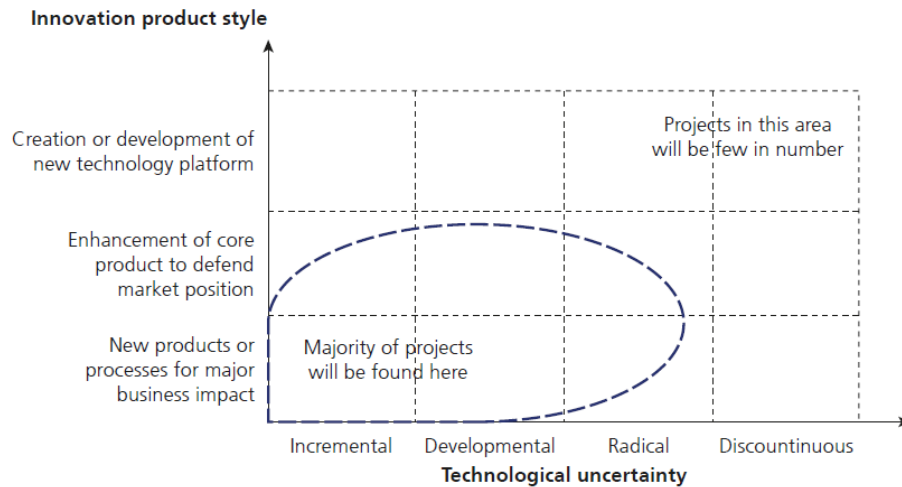


Figure 4.4 A two-dimensional typology of innovation projects

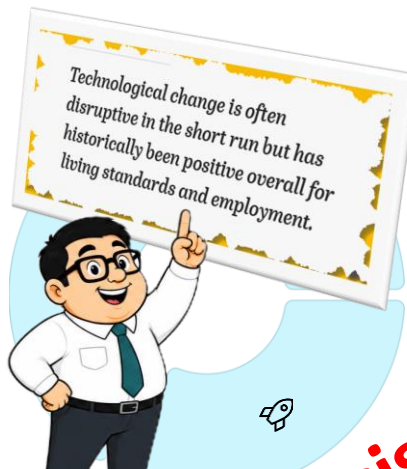
Source: P. Trott, Innovation Management and New Product Development, 7th Edition, Pearson Education, 2021.)

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Extent of impact caused by an Innovation

Progressive Innovation

Gradual improvements to existing products and services with relatively low risk and investment requirements.



Disruptive Innovation

Abrupt changes that transform markets, often starting in niche segments before overtaking established solutions.

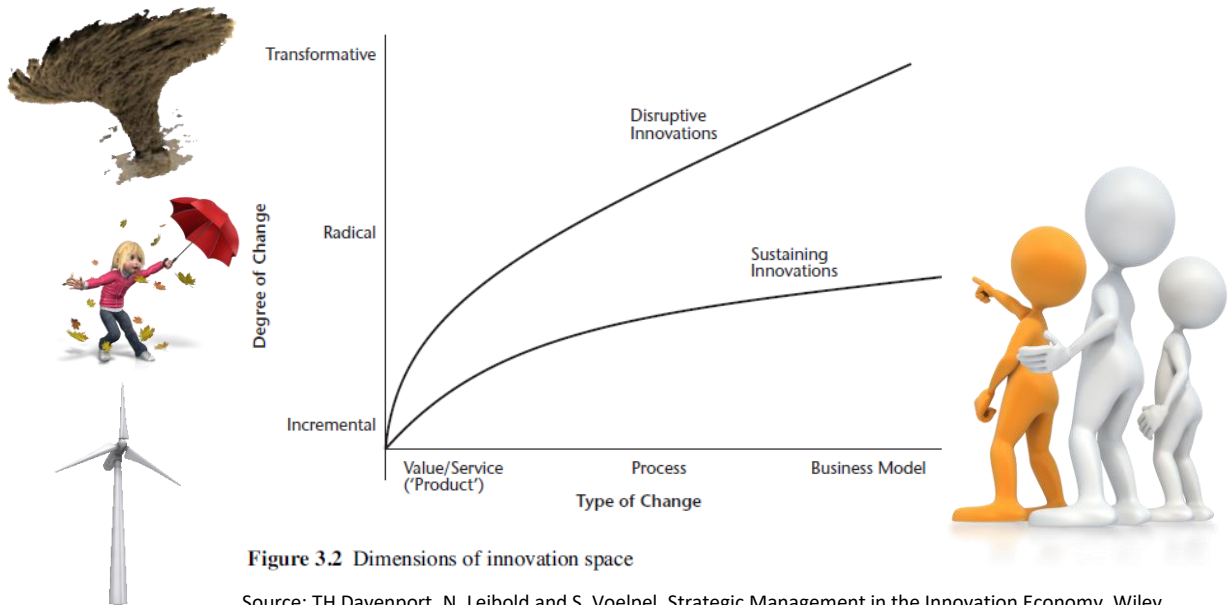
Radical Innovation

Revolutionary technologies that create entirely new needs, markets, and business models through breakthrough advancements.

Level of human resistance

446

“Change” associated with various types of innovations



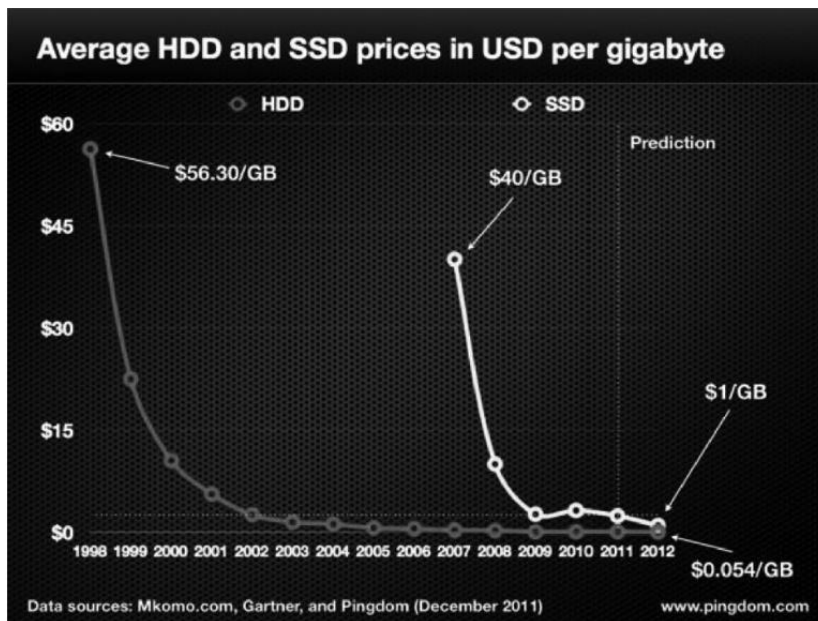
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Examples of How Changes in Environmental Trends Provided Openings for New Business and Product Opportunities

Changing Environmental Trend	Resulting New Business, Product, and Service Opportunities	Companies That Resulted
Economic Trends		
Rising personal incomes	Technology products	
Technological Advances		
Smartphones	Smartphone operating systems, smartphone apps, smartphone accessories	Android, Instagram, Snapchat, Rokit Boost, iSkin
Wearable technology	Wearable fitness devices, wearable lifestyle devices (like Google Glasses), and wearable tech gadgets	Apple Watch, Spectacles, Oculus Rift, Recon Jet
Political and Regulatory Changes		
Increased EPA and OSHA standards	Consulting companies, software to monitor compliance	MobileEpiphany, SWAC, ESS Group, Air Quality Consultants
Affordable Care Act	Electronic health records, new models of primary care, apps and technologies to deliver care more efficiently	CareCloud, VillageMD, Aledade, Ginger.io, Dispatch

Source: B. Barringer and R. Ireland (2019), Entrepreneurship Successfully launching new ventures, Pearson Education Limited, 2019.

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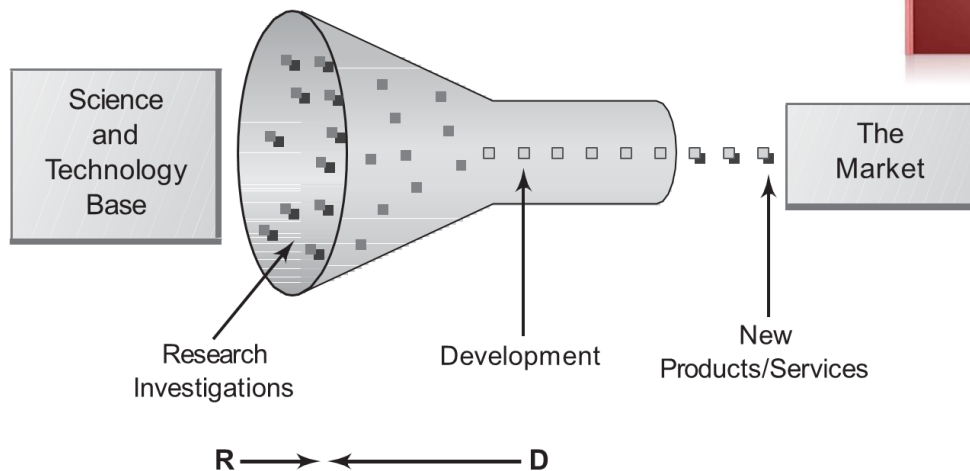
Source: H.J. Harriton (2019), Creativity, Innovation and Entrepreneurship, CRC Press, Taylor and Francis' Group, 2019.

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The Closed Innovation Paradigm

Source: Chesbrough, H. Open Services Innovation, 2011

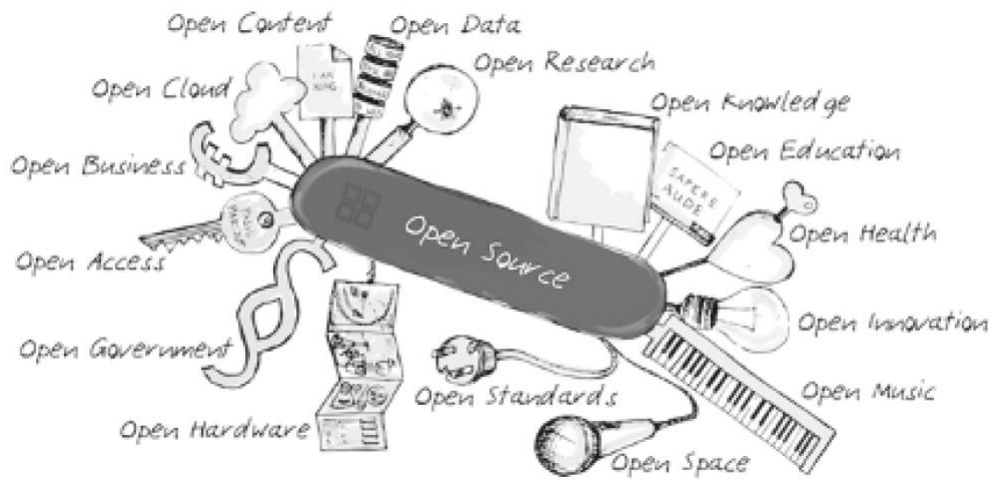
FIGURE 4.1 The Closed Innovation Paradigm



Source: © 2004 Henry Chesbrough.

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Benefits of an Open Source system



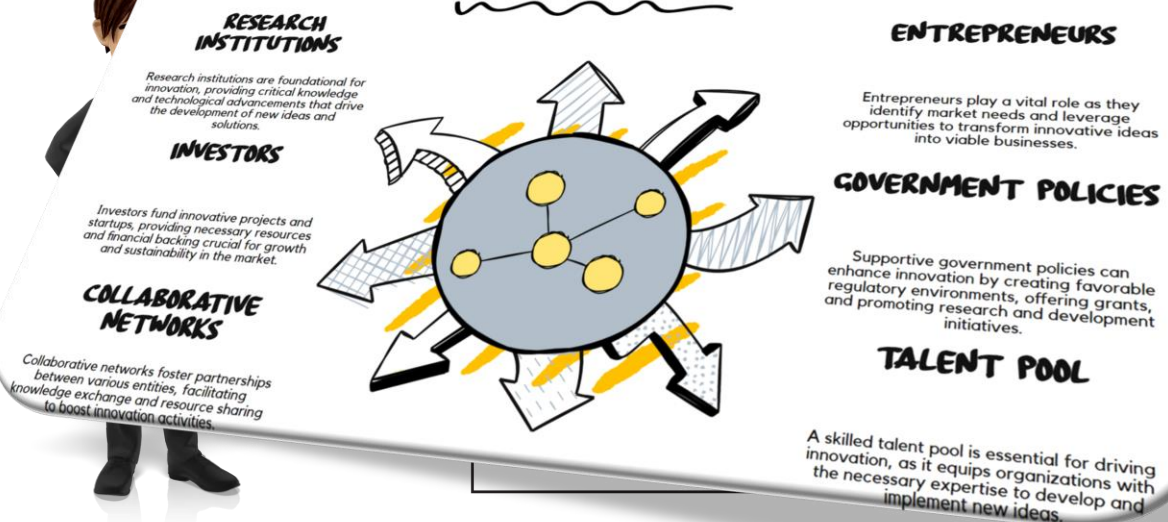
Source: C. Stimmel, *Evolving Innovation Ecosystems*, CRC Press 2017.

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External versus Internal Sourcing of Innovation

Key Elements of an Innovation Ecosystem

Understanding the core components driving innovation success.



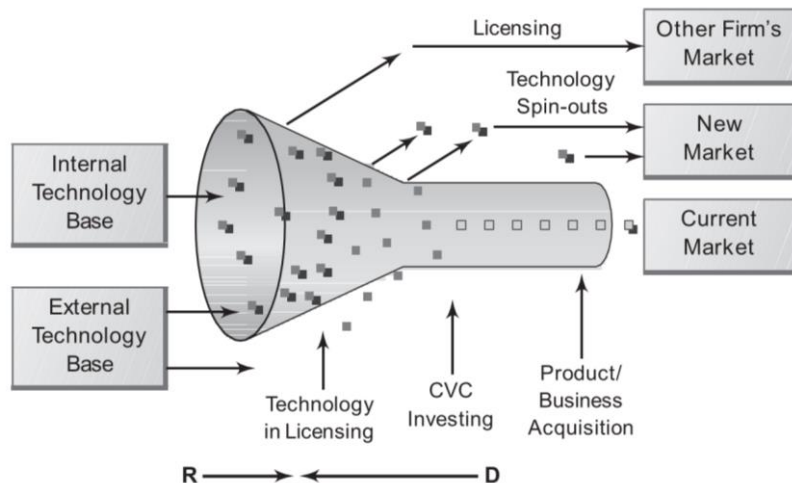
Source: M.A. Schilling, *Strategic Management of Technological Innovation*, 6th Edition, McGraw Hill, New York, 2020.

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The Open Innovation Paradigm

Source: Chesbrough, H. Open Services Innovation, 2011

FIGURE 4.2 The Open Innovation Paradigm



Source: © 2004 Henry Chesbrough.

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Approaches to innovation: Open vs Closed

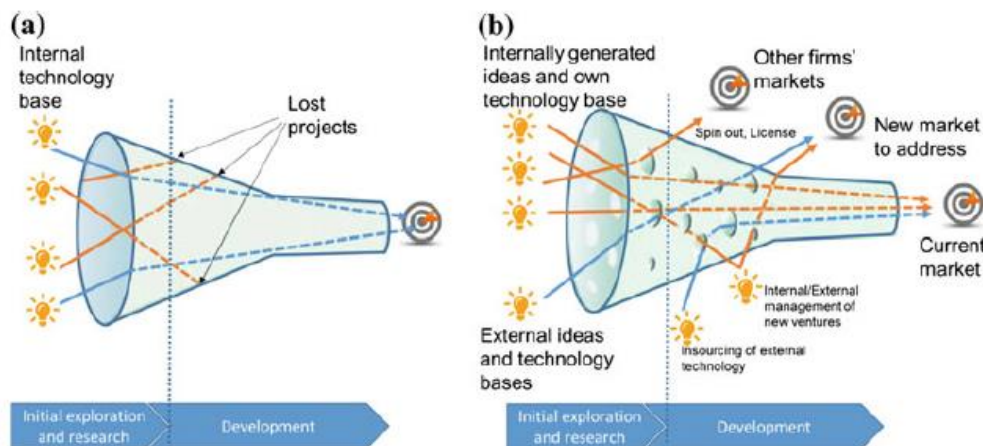


Fig. 8.6 The traditional (*closed funnel*) approach to innovation, compared to the Open Innovation (*open funnel*) approach

Source: M. Cantamessa and F. Montagna, Management of Innovation and Product Management, Springer-Verlag, London, 2016, 5

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ISE549 Management of Innovation and Technology

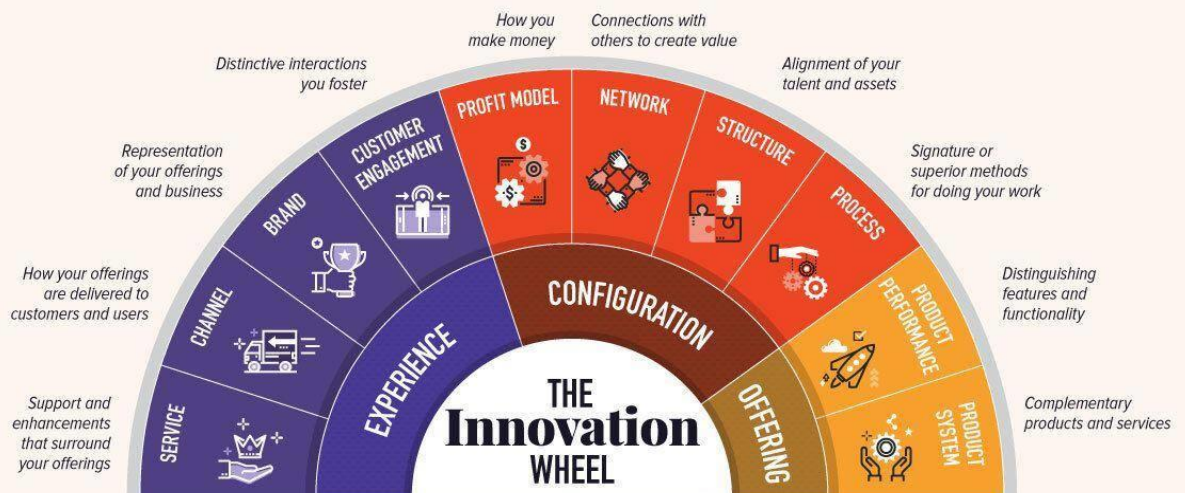
Case Studies

Prof. Eric Tsui

Educational Research & Technologies Centre
The Hong Kong Polytechnic University, Hong Kong

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UNDERSTANDING THE 10 Types of Innovation



<https://doblin.com/ten-types>

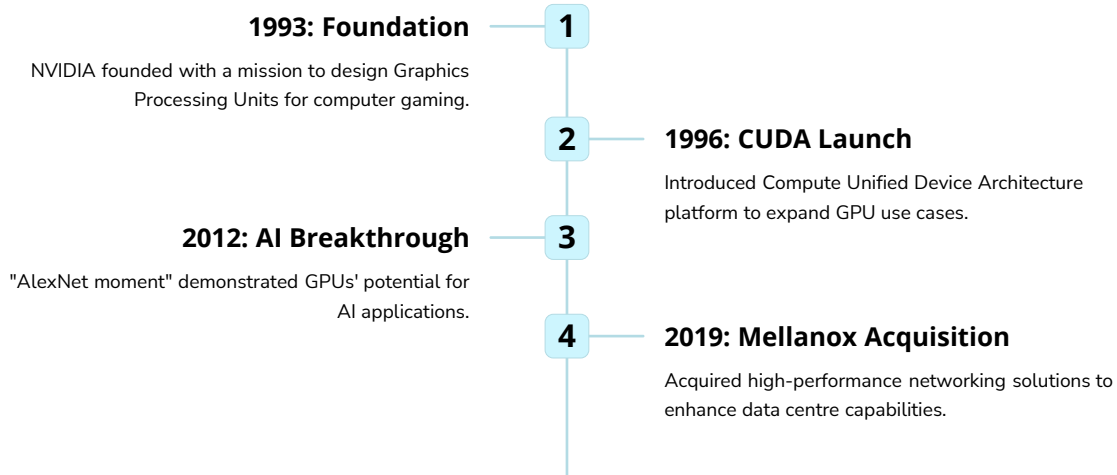
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NVIDIA: Harnessing Open Innovation to Promote User Lock-in

This presentation explores how NVIDIA built a complete digital technology stack. We'll examine how they combined proprietary and open innovation strategies to promote customer lock-in.

Source: <https://www.europeanbusinessreview.com/nvidia-harnessing-open-innovation-to-promote-user-lock-in/>

The NVIDIA Story



NVIDIA's Market Dominance

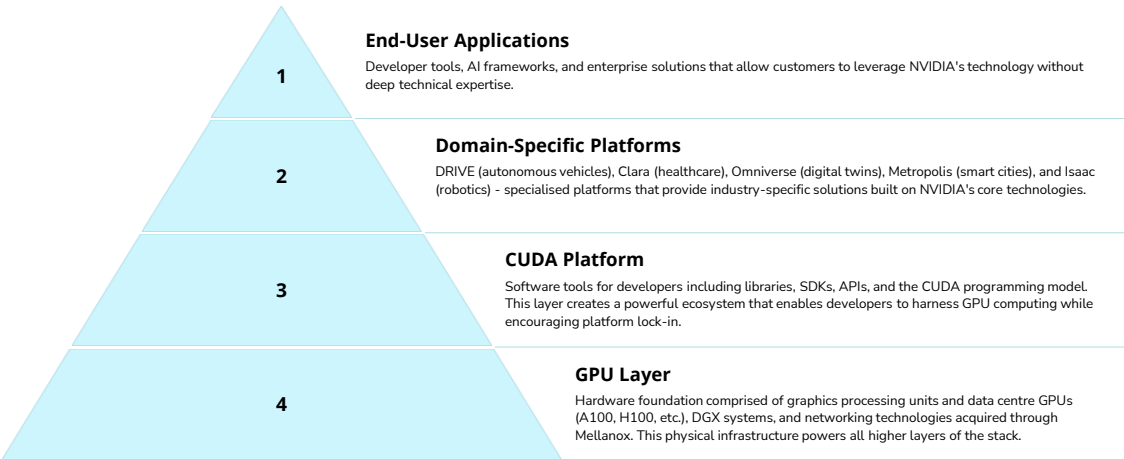
80%	84%	4M
AI Market Share	GPU Market Share	CUDA Developers
NVIDIA's high-end GPUs power approximately 80% of the generative AI market.	NVIDIA holds an impressive 84% of the overall GPU market.	Over 4 million developers use NVIDIA's CUDA platform.

Source: <https://www.europeanbusinessreview.com/nvidia-harnessing-open-innovation-to-promote-user-lock-in/>

3

NVIDIA's Technology Stack

NVIDIA has built a comprehensive vertical technology stack that creates significant barriers to entry for competitors while delivering powerful capabilities to users.



This integrated stack approach enables NVIDIA to maintain market dominance by providing end-to-end solutions while creating significant switching costs for users who adopt their technologies.

5

Open Innovation Model 1: Collaborative Innovation

NVIDIA strategically partners with industry leaders to enhance its ecosystem while maintaining proprietary advantages.

- 1

Controlled Co-Creation
Joint development of specific use cases with partners like Google and Meta. NVIDIA provides specialised hardware acceleration for their AI workloads while gaining insights into future technology needs.
 - Collaborative ventures with cloud providers (AWS, Google Cloud, Azure)
 - Strategic alliances with automotive manufacturers for DRIVE platform
 - Research partnerships with academic institutions that strengthen NVIDIA's talent pipeline
- 2

Framework Enhancement
Optimised TensorFlow and PyTorch for NVIDIA hardware. By contributing significant code to open-source AI frameworks, NVIDIA ensures superior performance on their GPUs compared to competitors.
 - Maintains dedicated teams working on framework optimisation
 - Provides cuDNN and NCCL libraries that accelerate deep learning operations
 - Creates framework-specific containers and SDKs that simplify deployment on NVIDIA infrastructure
- 3

Lock-In Strategy
Client systems become optimised for NVIDIA, increasing switching costs. Once organisations build workflows around NVIDIA's optimised software stack, migration to alternative hardware becomes technically challenging and expensive.
 - Implementation of proprietary extensions and optimisations in collaborative projects
 - Development of NVIDIA-specific features that don't translate to competitor hardware
 - Creation of dependency on NVIDIA's debugging and profiling tools within partner workflows

This approach allows NVIDIA to benefit from the innovation of partners while maintaining control of core technologies and strengthening market position.

Open Innovation Model 2: Accessible Resources

NVIDIA strategically provides free and accessible resources that simultaneously empower developers and create dependency on their ecosystem.

- 

CUDA Toolkit
Free access to development tools reduces entry barriers for new users. The toolkit includes libraries, debugging solutions, and compilers that optimise code for NVIDIA hardware, creating a path of least resistance toward NVIDIA adoption.
- 

Documentation
Comprehensive guides help developers master the CUDA ecosystem. These meticulously crafted resources focus exclusively on NVIDIA-optimised workflows, subtly discouraging exploration of competing platforms through omission.
- 

Educational Materials
Learning resources create developer dependence on NVIDIA's tools. From university partnerships to online courses, NVIDIA shapes the skills of emerging developers around their architecture, ensuring the next generation of AI specialists thinks in NVIDIA-first terms.
- 

Developer Community
NVIDIA fosters vibrant community forums and discussion groups where developers can find peer support. This community ecosystem reinforces NVIDIA-specific solutions while generating valuable user data and feedback that informs future product development.
- 

Sample Code & SDKs
Extensive libraries of ready-to-use code examples and specialised Software Development Kits simplify implementation of complex features. These resources dramatically accelerate development timelines while embedding NVIDIA-specific APIs throughout application codebases.
- 

Cloud Resources
NVIDIA provides access to GPU-accelerated cloud environments for testing and development. These platforms allow developers to experiment with high-performance computing without significant upfront investment, creating a smooth onramp to eventual hardware purchases.
- By providing these accessible resources, NVIDIA creates a powerful flywheel effect: more developers gain proficiency with their tools, leading to more NVIDIA-optimised applications, which increases hardware demand and strengthens their market position.
- 7
- 3

Open Innovation Model 3: Open-Source Resources

Selective Open-Sourcing

Only specific components like Linux kernel modules, drivers, and sample code repositories are made open source. The core platform and proprietary algorithms remain closed.

This strategic approach builds trust within the developer community while maintaining control of the ecosystem.

Examples include:

- CUDA-X libraries for specific domains
- NVIDIA Deep Learning Examples
- Container technologies and deployment tools

By contributing to open standards, NVIDIA shapes the direction of technology development while ensuring compatibility with their hardware.

Through these open-source initiatives, NVIDIA achieves the perfect balance: appearing generous with intellectual property while strategically reinforcing dependency on their hardware and ecosystem. Developers benefit from access to cutting-edge technology while becoming increasingly invested in NVIDIA's technical approach.

Recent Open-Source AI Models

NVLM 1.0 and Nemotron released under open-source licence in 2024, demonstrating NVIDIA's commitment to AI democratisation.

Performance rivals proprietary models from OpenAI and Google, while creating opportunities for researchers and businesses to innovate on NVIDIA platforms.

Key benefits include:

- Accelerated AI research on NVIDIA hardware
- User customisation for specific applications
- Expanded ecosystem of NVIDIA-optimised models
- Trust-building through transparency

These models perform best on NVIDIA GPUs, creating a natural hardware preference while appearing vendor-neutral.

Developer Community Engagement

NVIDIA maintains an active presence in open-source communities and foundations to shape standards and implementation practices.

Strategic contributions include:

- Supporting PyTorch, TensorFlow and other ML frameworks
- Contributing to standards like ONNX for model interoperability
- Sponsoring open-source events and hackathons
- Maintaining GitHub repositories with thousands of projects

This community engagement creates goodwill while ensuring NVIDIA's architecture remains the reference implementation for high-performance computing and AI workloads.

Open Innovation Strategies Summary

Strategy	Digital Affordance	Lock-In Effect
Collaborative Innovation	Controlled Co-Creation	Optimised for NVIDIA hardware
Openly Accessible Resources	Usability without Modification	Developer investment in CUDA skills
Open-Source Resources	Customisability	Integration into workflows

Key Takeaways

Strategic Balance

NVIDIA balances openness with proprietary control.

Innovation Model

A platform can share resources while creating a robust user base.



User Lock-In

Open innovation creates lasting dependence on NVIDIA's platform.

Market Dominance

Selective openness drives platform dominance and user loyalty.

Types of innovation

1. Incremental innovation
 2. Architectural innovation
 3. Modular innovation
 4. Radical innovation
 5. Disruptive innovation
- Process innovation
 - Service innovation
 - Open innovation
 - Business model innovation
 - Social innovation
 - Frugal innovation
 - Reverse innovation
 - Innovation during a crisis

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